

Trade Costs, Entry Costs, and the Spatial Distribution of Chinese Manufacturing Firms*

Zeyi Qian[†] Kensuke Suzuki[‡] Junfu Zhang[§]

July 23, 2025

Abstract

This paper examines sectoral growth patterns across Chinese provinces during the country's economic takeoff in the early 2000s, following key policy reforms including trade liberalization, infrastructure expansion, business climate improvements, and relaxed rural-to-urban migration restrictions. While coastal regions boomed as the economy integrated globally, particularly spurred by export-oriented processing plants, the significant growth occurring in inland regions during this same period is often overlooked. We develop a spatial general equilibrium model based on the Melitz-Chaney framework to analyze how these factors interact and influence the geographic distribution of economic activity. Our model extends the heterogeneous firm approach to a multi-sector, multi-region setting, incorporating detailed sectoral and regional linkages. Using Chinese data, we calibrate the model and conduct counterfactual simulations to identify the key mechanisms driving regional development. Our results highlight reductions in trade costs intensify competition and deter firm entry, improvements in the local business climate lower entry barriers and foster firm creation. Together, these factors shape China's overall economic growth and regional specialization. Building on this, We quantify the contributions of manufacturing employment expansion, trade cost reductions, and business climate improvements to overall welfare gains. Collectively, our decomposition of real wage growth reveals that selection effects dominate the growth of regional real wages, with these effects being more pronounced in inland regions compared to coastal regions.

*We thank Hayato Kato and Motoaki Takahashi for their valuable feedback during the early stages of this project. We also appreciate the insightful comments and constructive suggestions from seminar participants at University of Freiburg, Southeast University, Jinan University, Nankai University, Göttingen Workshop on International Economics, MEA Annual Conference, and CES Annual Conference. Any remaining errors are our own.

[†]Department of Economics, Clark University. Email: ZeQian@clarku.edu

[‡]Department of Economics, Clark University; Economic Research Center, Nagoya University. Email: KSuzuki@clarku.edu

[§]Department of Economics, Clark University. Email: JuZhang@clarku.edu

Keywords: Spatial Distribution of Firms, Trade Costs, Entry Costs, Melitz-Chaney Model, China’s Manufacturing

JEL Classification: F12, R12, L60

1 Introduction

China’s rapid economic transformation since the early 2000s is one of the most significant developments in modern economic history. As the country emerged as a global manufacturing hub, a growing body of research has explored the key drivers behind this shift. World Trade Organization (WTO) accession in 2001 and subsequent trade liberalization played a crucial role in integrating China into the global market and fostering industrial specialization (Lu and Yu, 2015; Brandt et al., 2017; Erten and Leight, 2021; Perla et al., 2021). Beyond reductions in international trade costs, large-scale infrastructure projects—such as the expansion of the highway network—have enhanced domestic market integration (Zheng and Kahn, 2013; Faber, 2014; Baum-Snow et al., 2017; Lin, 2017; Qin, 2017; Che et al., 2024). Additionally, labor market reforms, including the relaxation of rural-to-urban migration restrictions under the *Hukou* system (Au and Henderson, 2006; Fan, 2019; Tombe and Zhu, 2019; Ma and Tang, 2020; Fang and Huang, 2022; Imbert et al., 2022; Li et al., 2024; Zi, 2025), have contributed to the expansion of manufacturing employment. Another key factor is the increasingly business-friendly environment, characterized by streamlined administrative processes, tax incentives, and the establishment of special economic zones (Montinola et al., 1995; Li and Zhou, 2005; Zhang, 2011; Brandt et al., 2025). The relaxation of restrictions on private enterprises has further encouraged firm entry and growth.

Collectively, these changes have reshaped China’s industrial landscape, particularly the spatial distribution of firms and differential growth of productivity across regions. While much of the existing literature correctly emphasizes the rapid growth of coastal regions as export powerhouses, this perspective presents an incomplete picture, as it often overlooks the distinct economic growth and varied firm entry in the country’s inland regions (Zheng et al., 2014; Liu and Ma, 2023; Ma and Tang, 2024). Our paper challenges this prevailing narrative. A central finding of this research is that, contrary to common perception, China’s inland regions experienced surprisingly faster growth in manufacturing activity and productivity during key periods, even as coastal regions maintained their significant role.

This paper develops a unified quantitative framework to analyze the spatial distribution of manufacturing firms and its underlying mechanisms. The model builds on the general equilibrium framework of trade with heterogeneous firms *a la* Melitz-Chaney (Melitz, 2003; Chaney, 2008), extending it to incorporate multiple sectors (Caliendo and Parro, 2014) and multiple regions within a country (Redding, 2016; Redding and Rossi-Hansberg, 2017). This

structure allows us to measure the effects of changes in international and domestic trade costs, firm entry costs, and labor market frictions on the spatial distribution of economic activity. We quantify the model using detailed establishment-level data and interregional input-output tables from 2002 to 2007, a period of significant economic reforms. We calibrate key structural parameters and economic fundamentals, including national manufacturing employment, trade costs, and entry costs.

We conduct counterfactual analysis by solving the model in relative changes using exact hat algebra, following [Dekle et al. \(2008\)](#). Using 2002 as the base year, we introduce shocks to the model and simulate the economy under various counterfactual scenarios. First, we simulate the Chinese economy before the major reforms. Second, we assess the role of trade liberalization by adjusting international trade costs to their post-WTO accession levels and evaluate the effects of infrastructure development by setting domestic trade costs to their 2007 levels. Finally, we analyze the implications of an increasingly business-friendly environment by simulating the economy under the 2007 sunk entry costs for firms. Throughout these counterfactual exercises, we measure how changes in trade costs and entry costs shape the spatial distribution of economic activity and welfare.

Our analysis yields several fascinating insights into China’s economic transformation. We find that reductions in domestic trade costs tend to intensify import competition, which can in turn deter new firm entry. Conversely, lower entry costs foster firm entry by reducing barriers and contribute most significantly to overall welfare gains. Together, these factors shape China’s overall economic growth and regional specialization. We also follow [Caliendo et al. \(2023\)](#) and decompose real wage growth which reveals a striking pattern: selection effects dominate the growth of regional real wages, accounting for a substantial 67.2% of the increase. Moreover, these selection effects are significantly more pronounced in inland regions compared to coastal regions, challenges the prevailing perception of inland regions as lagging in China’s economic transformation.

This paper makes three key contributions to the literature. First, it develops a unified framework that integrates trade frictions, labor mobility, sectoral heterogeneity, and firms’ sunk entry costs to explain the spatial distribution of firms. Building on [Fujita et al. \(2001\)](#), who highlight how market integration influences firms’ location distribution, our model extends the insights of [Redding and Sturm \(2008\)](#) and [Tombe and Zhu \(2019\)](#) by incorporating both international and domestic trade barriers alongside labor mobility within a general equilibrium setting. This approach explicitly captures the interactions between these forces and highlights two key mechanisms through which the welfare effects of reducing trade costs are realized: intensified inter-regional competition, redistribution of firms, and differential growth of productivity across regions. Specifically, we show (1) how trade liberalization and infrastructure improvements lower trade costs and drive regional firm relocation, and (2) how regions exhibit distinct responses to these changes—for instance, our findings show that inland

regions experience stronger selection effects than coastal regions.

Second, this study sheds light on the role of regional characteristics—particularly the local business climate—in shaping firm location decisions. An influential early theory about the takeoff of the Chinese economy emphasizes “Chinese-style federalism,” where local governments spur economic growth by fostering entrepreneurship and attracting investment through development zones, tax breaks, subsidies, and low-cost land (Montinola et al., 1995; Li and Zhou, 2005). While recent research has largely focused on trade shocks and infrastructure development, such as WTO accession and improved transportation networks (e.g., Baldwin and Okubo, 2006; Allen and Arkolakis, 2014; Redding and Turner, 2015; Donaldson and Hornbeck, 2016; Henderson et al., 2018; Donaldson, 2018; Tombe and Zhu, 2019; Ma and Tang, 2020; Caliendo et al., 2021; Xu and Yang, 2021; Nagy, 2022; Balboni, 2025; Xu and Yang, 2025; Yuan and Ouyang, 2025), these two perspectives have almost always been studied in isolation. Our framework bridges this gap by integrating both views, examining how regional heterogeneity in business environment amplifies or mitigates the effects of broader policy changes. Crucially, our findings suggest that the contribution of reducing sunk entry costs plays a more important role than trade costs in shaping firm spatial distribution and regional development. This approach provides a more comprehensive understanding of the interplay between trade liberalization, infrastructure development, and regional disparities in business policies and promotions in China.

Third, our analysis highlights how sectoral specialization evolves in response to manufacturing expansion, declining trade costs, and lower entry barriers, crucially considering sectoral heterogeneity through factors like intermediate input linkages, amid China’s economic and policy shifts. We show that different sectors react uniquely to these changes, with regions becoming increasingly specialized as firms cluster in locations that offer comparative advantages. This finding aligns with Imbs and Wacziarg (2003), who emphasize sector-specific effects of trade liberalization, and Baum-Snow et al. (2017), who examine how infrastructure improvements drive firm relocation. Moreover, our model captures how these factors contribute to increasing specialization in manufacturing, reflecting mechanisms similar to those discussed by Glaeser et al. (1992), where economic growth is reinforced by the concentration of industries in favorable environments. By integrating these factors, our framework provides a richer understanding of how regional characteristics and policy shifts shape the evolving geography of sectoral specialization in China.

The remainder of the paper is structured as follows: Section 2 presents motivating evidence. Section 3 introduces the model framework. Section 4 quantifies the model and performs counterfactual analysis. Finally, Section 5 concludes.

2 Motivating Evidence

2.1 Development of China after 2000

Since the early 2000s, China’s economic transformation has been marked by significant shifts in the spatial distribution of economic activity. A key turning point was the country’s accession to the WTO in 2001, which accelerated its integration into the global economy. This integration had profound implications for trade and investment patterns across the country.

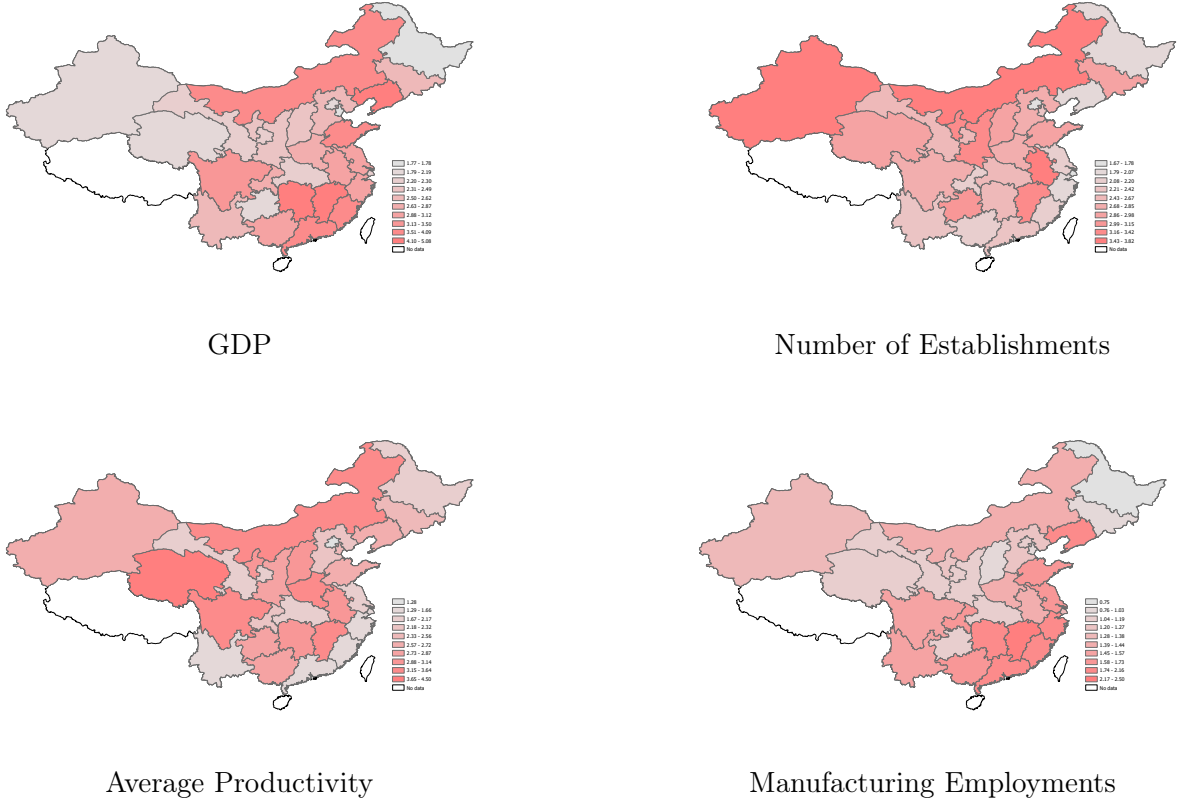
At the same time, China undertook extensive infrastructure development, particularly in high-speed rail and highway networks. Improved connectivity facilitated the movement of goods and services, allowing firms to expand their market reach. Between 2002 and 2007, international trade costs for non-agricultural sectors fell by 8%, while domestic trade costs declined by 11% (Tombe and Zhu, 2019). These trade cost reductions intensified market competition, forcing less productive firms to exit and enabling more competitive firms to expand, reshaping the geographic distribution of manufacturing activity.

Another major driver of economic reallocation was the relaxation of the *Hukou* system, which historically restricted rural-to-urban migration. Recent reforms have eased these restrictions, allowing more rural workers to move to urban areas and fueling growth in manufacturing employment. The resulting increase in labor supply enabled firms to scale production, reduce costs, and compete more effectively in both domestic and international markets.

Equally important, though often underestimated, is the role of regional development policies. In the early stages of economic reform, state-owned enterprises (SOEs) dominated the market, while private firms faced significant entry barriers due to strict regulations and limited access to resources. As reforms deepened and the business climate improved, these restrictions were gradually lifted, leading to a surge in private-sector participation. Local governments, operating within a system of *Chinese-style federalism*, actively promoted economic growth by establishing development zones, offering tax incentives, and providing subsidies and cheap land. Development zones, in particular, offered streamlined administrative support and well-integrated infrastructure, creating a business-friendly environment that encouraged entrepreneurship and investment.

The combined effects of trade liberalization, infrastructure expansion, rural-to-urban migration, and regional development policies led to a notable reallocation of economic activity across regions. Coastal provinces experienced rapid growth, driven by foreign direct investment and the establishment of export-oriented manufacturing plants. By 2007, the six largest exporting provinces—Guangdong, Fujian, Jiangsu, Shanghai, Shandong, and Zhejiang—were all in coastal areas, collectively accounting for 79.7% of China’s total exports. Figure 1 further highlights that the regions with the highest GDP growth from 2002 to 2007 were predominantly coastal, solidifying their role as key hubs in China’s manufacturing landscape.

Figure 1: Spatial Distribution of Manufacturing Sectors



Note: We aggregated firm-level data to calculate the number of establishments, regional GDP, average productivity, and manufacturing employments for both 2002 and 2007. The map visualizes the 2007 figures relative to the 2002 baseline, with darker shades representing higher values, highlighting regions with more significant growth.

While much of the focus has been on coastal growth, inland regions have also experienced significant industrial expansion. Since 2004, manufacturing output in these provinces has risen substantially (Zheng et al., 2014), driven by declining trade and entry costs, government-led initiatives such as the Western Development Program, and ongoing infrastructure improvements that enhanced market accessibility. Lower labor costs in these regions further incentivized firms to establish operations there. Figure 1 illustrates this shift, showing a growing concentration of manufacturing establishments in inland areas compared to coastal regions.

Concurrently, we are seeing a divergence: coastal areas are gaining more workers, while inland areas are becoming more productive. This occurs because easy knowledge diffusion reduces the need for workers to physically cluster, thus undermining the very productivity advantages that density is meant to create (Desmet et al., 2018).

2.2 Descriptive Evidence

To illustrate these trends, we first examine descriptive evidence on GDP growth and the expansion of establishments across different sectors. As shown in Table 1, between 2002 and 2007, manufacturing GDP grew by 211.06%, while the number of establishments increased by 122.88%, reflecting remarkable industrial expansion. Sectoral differences further highlight the complexity of China’s economic landscape. Among all industries, electronics and machinery manufacturing experienced the most rapid growth. These variations suggest that firms responded differently to changes in trade costs, rural-to-urban migration, and entry barriers, shaping the evolving distribution of economic activity.¹

To provide a more detailed representation of the spatial distribution of economic activity, we compute the index of specialization, which measures the extent to which an industry is concentrated in specific regions rather than evenly distributed across the country. Following [Ellison and Glaeser \(1997\)](#), we compute the specialization level of each sector as follows:

$$G^j = \sum_n \left(\frac{Y_n^j}{Y_C^j} - \frac{Y_n}{Y_C} \right)^2$$

where Y_n^j denotes the GDP of industry j in region n , Y_n represents the total GDP of region n , Y_C^j represents the total GDP in sector j , and Y_C is the overall GDP of China. A higher value of this measure indicates that an industry’s GDP is more heavily concentrated in certain regions rather than being evenly spread nationwide. Table 1 presents the specialization levels for each industry. By 2007, specialization had increased across all industries compared to 2002, indicating a more geographically concentrated distribution of economic activities.

These patterns highlight the evolving spatial distribution of economic activity in China and underscore the importance of understanding how factors such as trade liberalization, infrastructure expansion, local business climate, and labor mobility interact to shape the country’s manufacturing landscape.

3 Quantitative Model

To gain deeper insights into the spatial distribution of manufacturing firms, we build a model to examine the effects of factors such as trade costs, manufacturing employment, and firm entry costs. This approach allows us to identify the key drivers behind firm concentration in specific regions, providing a clearer understanding of how these forces shape the economic landscape.

¹In Appendix A, we also illustrate the spatial patterns of growth in two specific sectors (electronics and steel) and highlight two major corporations within these industries: Foxconn in electronics and Angang in steel.

Table 1: Growth in Output and Number of Establishments

	All Manufacturing	Food	Chemicals	Metals	Machinery	Electronics
	GDP (1 trillion CNY)					
2002	0.29	0.06	0.05	0.05	0.05	0.04
2007	0.91	0.19	0.14	0.15	0.16	0.15
Δ	0.62	0.13	0.09	0.10	0.11	0.10
$\%\Delta$	211.06%	198.50%	194.87%	222.48%	233.11%	241.09%
	Establishments (1k)					
2002	117.16	27.80	19.66	21.93	17.56	11.86
2007	261.12	59.95	42.60	46.28	46.56	28.62
Δ	143.96	32.15	22.94	24.36	29.00	16.76
$\%\Delta$	122.88%	115.66%	116.65%	111.07%	165.21%	141.36%
	Productivity - Value Added Per Capita (1K CNY)					
2002	72.76	68.18	87.01	65.99	66.55	94.05
2007	164.21	156.92	195.61	186.62	144.81	175.66
Δ	91.45	88.74	108.59	120.63	78.26	81.61
$\%\Delta$	125.69%	130.16%	124.80%	182.82%	117.58%	86.77%
	Sectoral Specialization Index (/100)					
2002		0.29	0.13	0.39	0.86	1.83
2007		0.48	0.17	0.40	0.055	2.65
Δ		0.19	0.04	0.01	-0.31	0.82
$\%\Delta$		63.91%	31.93%	1.21%	-36.77%	44.44%

Note: We consolidate the manufacturing industries into six sectors (5 specific sectors+1 other sector) and use *Chinese Industrial Enterprise Database* to calculate the outcomes, which will be discussed in detail in the calibration section. In Appendix B, we provide another way to measure specialization across regions by Herfindahl-Hirschman Index.

3.1 Environment

We consider a world consisting of two countries: China and the rest of the world (RoW). China is divided into N regions, which are represented by subscripts m and n ($= 1, 2, \dots, N$), while the RoW is denoted by subscript 0. For simplicity, the term “region” will refer to both Chinese regions and the RoW, with C used to refer to China and 0 for the RoW when specifying the countries. Labor is the only primary factor of production, and its endowment is fixed in each country. In China, labor is perfectly mobile both across regions and between rural and urban areas, allowing workers to move freely between agriculture and manufacturing. The economy includes J manufacturing sectors, indexed by superscripts i and $j = 1, 2, \dots, 6 \in J$ (or $M \in k$), and an agricultural sector, indexed by i and $j = 0 \in J$ (or $A \in k$).

3.2 Worker Preferences

In each country, there is an ex-ante identical continuum of consumer indexed by ω . The mass of workers in China and the RoW is fixed and represented by L_C and L_0 , respectively.

A worker ω 's preference for residing in the urban (manufacturing) or rural (agricultural) area k of the region n depends on the goods consumption index (C_n) and an idiosyncratic amenity shock (b_n^k), which affects their utility from living in that region:

$$u_n^k(\omega) = b_n^k(\omega) C_n(\omega). \quad (1)$$

As the RoW consists of a single region, we normalize $b_0^k(\omega) = 1$ for all worker ω . The goods consumption index C_n is defined as the Cobb-Douglas bundle of sectoral final goods C_n^j ,

$$C_n(\omega) = \prod_{j=0,1,\dots,6} \left(\frac{C_n^j(\omega)}{\alpha^j} \right)^{\alpha^j} \quad \text{with } \alpha^j \in (0,1) \text{ and } \sum_{j=0,1,\dots,6} \alpha^j = 1, \quad (2)$$

where $C_n^j(\omega)$ is worker ω 's consumption of final good j in region n and α^j is the expenditure share on final good j . From (1) and (2), we obtain the indirect utility of consumer ω residing in region n :

$$v_n^k(\omega) = b_n^k(\omega) \frac{e_n^k}{P_n}. \quad (3)$$

where $e_n(\omega)$ is the consumer's expenditure on goods and P_n is the consumer price index dual to (2),

$$P_n = \prod_{j=0,1,\dots,6} (P_n^j)^{\alpha^j}, \quad (4)$$

where P_n^j is the price of sectoral final good j in region n .

The idiosyncratic amenity shocks $b_n^k(\omega)$ capture heterogeneous preferences for living in urban or rural area of each region in China. Following Redding (2016), we assume that the shocks are drawn independently across Chinese regions and workers from a Fréchet distribution:

$$b_n^k(\omega) \sim G_n(b) = \exp\left(-B_n^k(\tilde{\gamma}b)^{-\eta}\right),$$

where the location parameter B_n^k determines average amenities for areas of region n and the shape parameter η governs the dispersion of amenities across individuals. $\tilde{\gamma} = \Gamma\left(1 - \frac{1}{\eta}\right)$ is a normalizing constant, where $\Gamma(\cdot)$ denotes the gamma function.

Workers choose the region-area pair that maximizes their utility after observing the

realization of idiosyncratic amenity shocks. The optimization problem is given by,

$$\max_{n,k \in \{A,M\}} v_n^k(\omega) = b_n^k(\omega) \frac{e_n^k}{P_n}.$$

A worker forms an expectation of expenditure $e_n(\omega)$ when choosing between the region-area pairs. This is because the specific manufacturing sectors in which the worker will be employed is not known at the time of choosing to reside in the urban sector. In contrast, the rural sector consists of a single agricultural activity with no such uncertainty.

3.3 Production Technology

We model the economy with two segments that differ in their structural assumptions: agriculture, featuring homogeneous output and a representative production technology, and manufacturing, characterized by firm heterogeneity and monopolistic competition.

3.3.1 Agriculture

Agricultural output in region n is given by a Cobb-Douglas function of labor and land:

$$y_n^A = \mathcal{A}_n^A (L_n^A)^{\beta^A} (D_n)^{1-\beta^A},$$

where $\mathcal{A}_n^A$ denotes exogenous agricultural productivity, L_n^A is the labor employed in agriculture, and D_n is the fixed regional land endowment.

The agricultural price index p_n^A is derived from the cost-minimization problem of a representative producer and is given by the unit cost function:

$$P_n^A = \frac{(w_n^A)^{\beta^A} (\tilde{r}_n)^{1-\beta^A}}{\mathcal{A}_n^A},$$

where w_n^A is the wage for agricultural workers, $\tilde{r}_n$ is the rental rate of land, and $\mathcal{A}_n^A$ is total factor productivity in agriculture.

3.3.2 Manufacturing

Intermediate Good Producers

Each manufacturing sector features a continuum of firms producing unique varieties under monopolistic competition. Firms are heterogeneous in their core productivity *à la* Melitz (2003). Let φ_n^j be the productivity of a firm in sector j located in region n . We will use the firm's productivity as an index of the firm (variety). The Production function of a firm φ_n^j in

sector j based at region n is given by,

$$y_n^j(\varphi_n^j) = \varphi_n^j \left(\frac{\ell_n^j(\varphi_n^j)}{\beta^j} \right)^{\beta^j} \left(\frac{m_n^j(\varphi_n^j)}{1 - \beta^j} \right)^{1 - \beta^j},$$

where $\ell_n^j(\varphi_n^j)$ is labor input in terms of efficiency unit, $m_n^j(\varphi_n^j)$ is the composite intermediate inputs, and β^j is the cost share of labor input in total production cost. Following the sectoral input-output linkage formulated by [Caliendo and Parro \(2014\)](#), we assume that firms use sectoral final goods (defined below) from all sectors as intermediate inputs, such that,

$$m_n^j(\varphi_n^j) = \prod_{i \in \{1, 2, \dots, J\}} \left(\frac{m_n^{ji}(\varphi_n^j)}{\gamma^{ji}} \right)^{\gamma^{ji}}, \quad (5)$$

where $m_n^{ji}(\varphi_n^j)$ is the quantity of final good i used to produce variety φ_n^j and γ^{ji} is the intermediate input elasticity related.

Solving the cost minimization problem of a firm gives the marginal cost of production as $c_n^j(\varphi_n^j) = \frac{c_n^j}{\varphi_n^j}$, where c_n^j is the cost of input bundle defined as,

$$c_n^j = (w_n^M)^{\beta^j} (\Xi_n^j)^{1 - \beta^j}, \quad (6)$$

where w_n^M is the average wage for manufacturing, and Ξ_n^j is the composite price index of intermediate inputs, dual to (5),

$$\Xi_n^j = \prod_{i \in \{1, 2, \dots, J\}} (P_n^i)^{\gamma^{ji}}, \quad (7)$$

where P_n^i is the price index of final good i .

A firm in sector j based in region n needs to incur two types of costs to serve the market in region m . The first cost is the variable iceberg trade costs, and the second is a fixed cost, which will be introduced below. To ship one unit of sector j good from n to m , a firm needs to ship τ_{mn}^j units of the goods. $\tau_{mn}^j \geq 1$ and we assume $\tau_{nn}^j = 1$ for all m, n and the triangular inequality holds. The marginal cost of serving goods from n to m net of iceberg trade costs is $c_{mn}^j(\varphi_n^j) = \frac{c_{mn}^j}{\varphi_n^j}$, where

$$c_{mn}^j = \tau_{mn}^j c_n^j. \quad (8)$$

Final Good Producers

In each sector and region, there is a final good producer that purchases and aggregates all of the varieties sold to the market. As in [Armington \(1969\)](#) and [Caliendo et al. \(2023\)](#), we

consider the nested CES aggregator and to avoid corner solutions in multi-sector monopolistic competition models, we follow [Kucheryavyy et al. \(2023\)](#) and adopt a three-tier CES preference structure. The upper-tier is the aggregation between goods sourced from China and RoW,

$$Q_n^j = \left(\left(\tilde{Q}_{nC}^j \right)^{\frac{\rho^j-1}{\rho^j}} + \left(\tilde{Q}_{n0}^j \right)^{\frac{\rho^j-1}{\rho^j}} \right)^{\frac{\rho^j}{\rho^j-1}}. \quad (9)$$

where $\tilde{Q}_{nC}^j$ denotes the aggregation among goods sourced from different Chinese regions in sector j at destination n , as the middle-tier,

$$\tilde{Q}_{nC}^j = \left(\sum_{m \in \{1,2,\dots,N\}} \left(y_{nm}^j(\phi) \right)^{\frac{\epsilon^j-1}{\epsilon^j}} \right)^{\frac{\epsilon^j}{\epsilon^j-1}},$$

and $\tilde{Q}_{n0}^j$ is the composite of RoW-sourced varieties. The lower-tier is the aggregation in a Chinese region m among different varieties sold to region n ,

$$y_{nm}^j(\phi) = \left(\int_{\phi \in \Omega_{nm}^j} \left(q_{nm}^j(\phi) \right)^{\frac{\sigma^j-1}{\sigma^j}} d\phi \right)^{\frac{\sigma^j}{\sigma^j-1}},$$

where $q_{nm}^j(\phi)$ is the demand for variety ϕ produced in region m and sold in n . σ^j denotes the elasticity of substitution across varieties produced within a region (lower-tier CES elasticity), while ϵ^j represents the elasticity of substitution across goods produced within a country (middle-tier CES elasticity), and ρ^j substitution between domestic and foreign-produced goods (upper-tier CES elasticity). Ω_{nm}^j is the set of varieties produced in m and sold in n , which is determined in equilibrium.

By solving a firm's profit maximization problem, we get the firm's pricing rule,

$$p_{nm}^j(\varphi_m^j) = \mu^j c_{nm}^j(\varphi_m^j), \quad (10)$$

where $\mu^j = \frac{\sigma^j}{\sigma^j-1}$ is markup for sector j . Denoting $r_{nm}(\varphi_m^j)$ as the firm's revenue, we can show that the firm's operating profit is a constant share of revenue,

$$\begin{aligned} \varpi_{nm}^j(\varphi_m^j) &= r_{nm}^j(\varphi_m^j) - q_{nm}^j(\varphi_m^j) c_{nm}^j(\varphi_m^j) \\ &= q_{nm}^j(\varphi_m^j) p_{nm}^j(\varphi_m^j) - q_{nm}^j(\varphi_m^j) c_{nm}^j(\varphi_m^j) \\ &= \frac{1}{\sigma^j} r_{nm}^j(\varphi_m^j). \end{aligned}$$

Price index of the sectoral final good dual to (9) is given by

$$P_n^j = \left(\left(\tilde{P}_{nC}^j \right)^{1-\rho^j} + \left(\tilde{P}_{n0}^j \right)^{1-\rho^j} \right)^{\frac{1}{1-\rho^j}}, \quad (11)$$

where $\tilde{P}_{nC}^j$ and $\tilde{P}_{n0}^j$ are, respectively, given by,

$$\tilde{P}_{nC}^j = \left(\sum_{m \in \{1,2,\dots,N\}} \left(\int_{\phi \in \Omega_{nm}^j} (p_{nm}^j(\phi))^{1-\sigma^j} d\phi \right)^{\frac{1-\epsilon^j}{1-\sigma^j}} \right)^{\frac{1}{1-\epsilon^j}}, \quad (12)$$

and

$$\tilde{P}_{n0}^j = \left(\int_{\phi \in \Omega_{n0}^j} (p_{n0}^j(\phi))^{1-\sigma^j} d\phi \right)^{\frac{1}{1-\sigma^j}}.$$

The nested CES structure of the final good aggregator implies that the expenditure share on goods sourced from Chinese regions is given by,

$$g_{nC}^j = \frac{\left(\tilde{P}_{nC}^j \right)^{1-\rho^j}}{\left(P_n^j \right)^{1-\rho^j}}, \quad (13)$$

and the share on RoW-sourced goods is,

$$g_{n0}^j = \frac{\left(\tilde{P}_{n0}^j \right)^{1-\rho^j}}{\left(P_n^j \right)^{1-\rho^j}}, \quad (14)$$

Furthermore, we can show that the expenditure share on goods sourced from a Chinese region m in total spending on China-sourced goods is

$$\lambda_{nm}^j = \frac{\left(\zeta_{nm}^j \right)^{1-\epsilon^j}}{\left(\tilde{P}_{nC}^j \right)^{1-\epsilon^j}}, \quad (15)$$

where

$$\zeta_{nm}^j = \left(\int_{\phi \in \Omega_{nm}^j} (p_{nm}^j(\phi))^{1-\sigma^j} d\phi \right)^{\frac{1}{1-\sigma^j}}, \quad (16)$$

for $n \in \{0, 1, \dots, N\}$ and $m \neq 0$. Using ζ_{nm}^j , we can re-express (12) as,

$$\tilde{P}_{nC}^j = \left(\sum_{m \in \{1, 2, \dots, N\}} (\zeta_{nm}^j)^{1-\epsilon^j} \right)^{\frac{1}{1-\epsilon^j}}. \quad (17)$$

Sunk Entry Costs and Fixed Costs to Serve Each Market

Firms entering the market in sector j in region m must pay a sunk entry cost to draw its productivity, measured as f_m^{jE} efficiency units of labor from the local market of sector j . Moreover, to serve in market n , firms need to pay a fixed cost, again in terms of efficiency units of local labor, f_{nm}^j . Fixed cost of serving the home market is presumably lower than the ones of serving the other regions, i.e., $f_{nm}^j \geq f_{mm}^j$. A firm's profit from serving market m net of fixed cost is given by,

$$\pi_{nm}(\varphi_m^j) = \varpi_{nm}^j(\varphi_m^j) - f_{nm}^j w_m^M.$$

Since $\pi_{nm}(\varphi_m^j)$ is monotonically increasing in productivity φ_m^j , there is a unique productivity cutoff ϕ_{nm}^j , which indicates the productivity of the least productive firms serving from m to n , defined by

$$\pi_{nm}(\phi_{nm}^j) = 0. \quad (18)$$

Firm's profit net of sunk entry cost is given by,

$$\sum_{n \in \{0, 1, 2, \dots, N\}} \pi_{nm}(\varphi_m^j) - f_m^{jE} w_m^M.$$

Here, the sunk entry cost captures the initial investment a firm must make to enter the market, including the cost of hiring efficiency units of labor in the local market. It is a one-time cost that is independent of future production and sales. In addition to labor costs, it reflects factors like local infrastructure, regulations, and the overall business environment. These costs act as barriers to entry, as only firms with sufficient productivity can overcome them and enter the market.

3.4 Aggregation and General Equilibrium

Price Indices and Goods Market Clearing

Following Chaney (2008), we assume that ex-ante identical entrants draw productivity from a Pareto distribution with the cumulative distribution function $G_n^j(\phi) = \Pr(\varphi_n^j < \phi) = 1 - \left(\frac{\tilde{b}_n^j}{\phi}\right)^{\theta^j}$ and the probability density function $g_n^j(\phi) = \frac{d}{d\phi} G_n^j(\phi)$ where $\varphi_n^j \geq \tilde{b}_n^j$. The shape

parameter θ^j differs across sectors. The location parameter $\tilde{b}_n^j$ differs across sectors and regions. Let T_n^j be the mass of entrants, determined by the free-entry condition in equilibrium. Sales of firm φ_m^j based in region m selling to region n can be expressed as:

$$r_{nm}^j(\varphi_m^j) = \begin{cases} X_{nC}^j \left(\frac{p_{nm}^j(\varphi_m^j)}{\zeta_{nm}^j} \right)^{1-\sigma^j} \left(\frac{\zeta_{nm}^j}{\tilde{P}_{nC}^j} \right)^{1-\epsilon^j} & \text{for } m \neq 0 \\ X_{n0}^j \left(\frac{p_{nm}^j(\varphi_m^j)}{\tilde{P}_{n0}^j} \right)^{1-\sigma^j} & \text{for } m = 0 \end{cases},$$

where X_{nC}^j is the total expenditure on China-sourced goods in region n and X_{n0}^j is the total expenditure on the ROW-sourced goods. From (18), the zero cutoff profit condition defines the cutoff productivity above which firms serve market in region n from region m is given by,

$$\phi_{nm}^j = \begin{cases} \left(\frac{\sigma^j w_m^M f_{nm}^j (\mu^j c_{nm}^j)^{\sigma^j-1}}{X_{nC}^j (\zeta_{nm}^j)^{\sigma^j-\epsilon^j} (\tilde{P}_{nC}^j)^{\epsilon^j-1}} \right)^{\frac{1}{\sigma^j-1}} & \text{for } m \neq 0 \\ \left(\frac{\sigma^j w_0^M f_{n0}^j (\mu^j c_{n0}^j)^{\sigma^j-1}}{X_{n0}^j (\tilde{P}_{n0}^j)^{\sigma^j-1}} \right)^{\frac{1}{\sigma^j-1}} & \text{for } m = 0 \end{cases}. \quad (19)$$

In our model, competition affects the selection of firms through the productivity cutoff ϕ_{nm}^j , which determines which firms can profitably enter and serve a market. Wage w_m^M , fixed costs f_{nm}^j , and production costs c_{nm}^j raise the cutoff productivity required for firms to serve. In addition, higher expenditures and a lower price index in region n reduce the cutoff. These factors drive a competitive selection effect, where only firms with higher productivity can survive and compete, while less productive firms are forced to exit the market.

Using the cutoff productivity defined by (19), we define the average productivity of firms selling goods from m to n as,

$$\begin{aligned} \bar{\phi}_{mn}^j &= \left(\frac{1}{1 - G_n^j(\phi_{mn}^j)} \int_{\phi_{mn}^j} (\phi)^{\sigma^j-1} dG^j(\phi) \right)^{\frac{1}{\sigma^j-1}} \\ &= \left(\frac{\theta^j}{\theta^j + 1 - \sigma^j} (\phi_{mn}^j)^{\sigma^j-1} \right)^{\frac{1}{\sigma^j-1}}. \end{aligned} \quad (20)$$

Furthermore, the mass of firms selling from n to m is given by

$$T_{nm}^j = (1 - G_n^j(\phi_{nm}^j)) T_m^j = \left(\frac{\tilde{b}_m^j}{\phi_{nm}^j} \right)^{\theta^j} T_m^j. \quad (21)$$

Combining equations (10), (16), (20), and (21), we can solve for the price index of goods

sourced from Chinese region m sold in region n as

$$\zeta_{nm}^j = \frac{\mu^j \left(T_{nm}^j\right)^{\frac{1}{1-\sigma^j}} c_{nm}^j}{\bar{\phi}_{nm}^j} \quad \text{for } m \neq 0, \quad (22)$$

and the price index of goods sourced from the RoW and sold in region n as

$$\tilde{P}_{n0}^j = \frac{\mu^j \left(T_{n0}^j\right)^{\frac{1}{1-\sigma^j}} c_{n0}^j}{\bar{\phi}_{n0}^j}. \quad (23)$$

Spending on final goods j is the sum of expenditure by workers on it as final consumption and by firms on it as an intermediate input. Let HE_n^j be the worker expenditure and ME_n^j be the firm expenditure. Total spending on final good j is given by

$$X_n^j = HE_n^j + ME_n^j. \quad (24)$$

Spending on China-sourced goods in region n is given by

$$X_{nC}^j = g_{nC}^j X_n^j, \quad (25)$$

where g_C^j is the expenditure share on China-sourced goods in total spending on final goods defined by (13). Analogously, we can express the spending on the RoW-sourced goods as

$$X_{n0}^j = (1 - g_{nC}^j) X_n^j. \quad (26)$$

The bilateral trade flow, i.e., the total spending on goods sourced from m and sold in n can be expressed as

$$X_{nm}^j = \begin{cases} \lambda_{nm}^j X_{nC}^j & \text{for } m \neq 0 \\ X_{n0}^j & \text{for } m = 0 \end{cases}. \quad (27)$$

The goods market clearing condition, expressed as the total sales of firms selling from region m being equal to the sum of total spending on the goods over all destination regions, is given by

$$Y_n^j = \sum_{m=0,1,\dots,N} X_{mn}^j. \quad (28)$$

And finally, intermediate input demand can be expressed using the aggregate sales Y_n^j as

$$ME_n^j = \sum_{i=1,2,\dots,6}^J \frac{Y_n^i}{\mu^i} (1 - \beta^i) \gamma^{ij}. \quad (29)$$

Free Entry of Firms

We assume that firms enter the market until the ex-ante expected profit is zero, i.e.,

$$\mathbb{E} \left[\sum_{n \in \{0,1,2,\dots,N\}} \pi_{nm}(\varphi_m^j) \right] = f_m^{jE} w_m^M,$$

where the expectation is taken over the productivity. This suggests that, in equilibrium, the aggregate profit will be zero. To derive the free entry condition determining the mass of entrants T_m^j , we first show that the total spending on fixed costs by all firms serving from region m to n , F_{nm}^j , is a constant fraction of the aggregate sales, i.e.,

$$\begin{aligned} F_{nm}^j &= T_{nm}^j w_m^M f_{nm}^j \\ &= \frac{\theta^j + 1 - \sigma^j}{\sigma^j \theta^j} X_{nm}^j. \end{aligned} \quad (30)$$

Since the operating profit of the firm is a constant share of revenue, by combining the result from (30), we can show that the total profit of firms producing in region m , net of fixed cost payment, is also a constant share of total sales, i.e.,

$$\begin{aligned} \Pi_m^j &= \sum_{n \in \{0,1,2,\dots\}} \left(\int_{\phi_{nm}^j}^{\infty} \pi_{nm}^j(\varphi) g_m^j(\varphi) d\varphi \right) \\ &= \sum_{n \in \{0,1,2,\dots\}} \left(\frac{1}{\sigma^j} X_{nm}^j - F_{nm}^j \right) \\ &= \left(\frac{\sigma^j - 1}{\sigma^j \theta^j} \right) Y_m^j. \end{aligned}$$

Free entry of firms in equilibrium implies that the total profit of firms in the region m (net of fixed costs) is equal to the payment for the sunk entry cost: $\left(\frac{\sigma^j - 1}{\sigma^j \theta^j} \right) Y_m^j = T_m^j w_m^M f_m^{jE}$. This gives the mass of entrants entering region m in sector j ,

$$T_m^j = \frac{\left(\frac{\sigma^j - 1}{\sigma^j \theta^j} \right) Y_m^j}{w_m^M f_m^{jE}}. \quad (31)$$

Thus the number of entrants T_m^j , is inversely proportional to sunk entry costs f_m^{jE} , meaning that higher sunk entry costs lead to fewer entrants.

Rural-Urban Sorting and Labor Market Clearing

We now explain the aggregate results for workers' rural-urban sorting backward. Given the real wage, by taking advantage of Fréchet distributed amenity shock, we can derive the probability that a randomly chosen worker chooses region n and sector k as follows,

$$\psi_n^k = \begin{cases} \frac{B_n^M \left(\frac{w_n^M}{P_n} \right)^\eta}{\sum_m B_m^k \left(\frac{w_m^k}{P_m} \right)^\eta} & \text{for } k = M \\ \frac{B_n^A \left(\frac{w_n^A}{P_n} \right)^\eta}{\sum_m B_m^k \left(\frac{w_m^k}{P_m} \right)^\eta} & \text{for } k = A \end{cases}. \quad (32)$$

And by the law of large numbers, we have the measure of workers residing in region n and area k in China is

$$L_n^k = \begin{cases} \psi_n^M L_C & \text{for } k = M, n \neq 0 \\ \psi_n^A L_C & \text{for } k = A, n \neq 0 \end{cases}, \quad (33)$$

where L_C is the measure of workers in China, which is exogenously given in the model.

Since wage is the only source of income for workers, their expenditure can be expressed as

$$E_n = L_n^M w_n^M + L_n^A w_n^A, \quad (34)$$

and sectoral spending is given by

$$E_n^j = \alpha^j E_n. \quad (35)$$

We then show the labor market clearing condition for manufacturing sectors,

$$\begin{aligned} L_n^M w_n^M &= \sum_{j=1,2,\dots,6} \left(\beta^j \frac{Y_n^j}{\mu^j} + \frac{1}{\sigma^j} Y_n^j \right) \\ &= \sum_{j=1,2,\dots,6} \left(\frac{\beta^j (\sigma^j - 1) + 1}{\sigma^j} \right) Y_n^j, \end{aligned} \quad (36)$$

where the left-hand side of the equation is labor supply in terms of value, and the right-hand side is the labor demand, which is the sum of demand for labor as variable costs and as fixed

and sunk entry costs. And the labor market clearing condition for agricultural sector is

$$\begin{aligned} L_n^A w_n^A &= \beta^A Y_n^A \\ &= \beta^A X_n^A \\ &= \beta^A E_n^A. \end{aligned} \tag{37}$$

Finally, the land market clearing condition is trivial in this setting, as agricultural production is the sole user of land. Thus, land used in agriculture equals the regional land endowment:

$$\begin{aligned} D_n \tilde{r}_n &= (1 - \beta^A) Y_n^A \\ &= (1 - \beta^A) X_n^A \\ &= (1 - \beta^A) E_n^A. \end{aligned} \tag{38}$$

Welfare

Following Redding (2016), the welfare is represented by the expected utility for a worker across regions, which is given by the following expression:

$$W_C = \bar{U} = \left(\sum_n \sum_k B_n^k \left(\frac{w_n^k}{P_n} \right)^\eta \right)^{1/\eta}, \tag{39}$$

where $\frac{w_n^k}{P_n}$ represents real wage. A higher real wage indicates that workers can purchase more goods with their income, leading to an increase in welfare. Consequently, regions with higher real wages contribute more significantly to overall welfare.

General Equilibrium

A general equilibrium of the model is a vector of wages $\{w_n^k\}_{n=0,\dots,N}^k$, price indices $\{\tilde{P}_{n0}^j\}_{n=0,\dots,N}^j$, $\{\tilde{P}_{nC}^j\}_{n=0,\dots,N}^j$, labor allocation $\{L_n^k\}_{n=0,\dots,N}^k$, measure of entrants $\{T_n^j\}_{n=0,\dots,N}^j$, and cutoff productivity $\{\phi_{mn}^j\}_{m,n=0,\dots,N}^j$ that satisfy the following equilibrium conditions: (i) *utility maximization by each household*, which yields price index (4); (ii) *profit maximization by each firm*, implying (10) and (11); (iii) *zero profit of the marginal firm as a result of free entry*, which determines the cutoff productivity and measure of entrants in (19) and (31); (iv) *goods markets clearing for every sector in each region*: (25) and (26); and (v) *labor market clearing for every sector in each region*, i.e., (36).

3.5 Equilibrium in Relative Changes

Our research employs a two-stage approach for quantification below. First, we solve the model in its level form. Despite data limitations that preclude the full calibration of fixed costs (f_{nm}^j) and the Pareto location parameter ($\tilde{b}_n^j$), we find that the model fits the observed data perfectly (see Appendix H). Second, to circumvent the challenge of directly calibrating these aforementioned parameters—namely, the Pareto location parameter and fixed costs—we employ the exact “hat algebra” method, in the spirit of Dekle et al. (2008). This approach allows us to directly conduct the counterfactual exercises while circumventing the calibration of the fundamentals. For a generic variable x , we define the “hat” notation,

$$\hat{x} = \frac{x'}{x},$$

where x is the variable in the baseline equilibrium and x' is the one in the counterfactual equilibrium.

For given structural parameters, θ^j , $\tilde{b}_n^j$, σ^j, ϵ^j , η , α^j , β^j , γ^{ji} , and ρ^j , and for given fundamentals in relative changes, $\{\hat{L}_C\}$, $\{\hat{L}_0\}$, $\{\hat{D}_n\}$, $\{\hat{\tau}_{nm}^j\}$, $\{\hat{B}_n^j\}$, $\{\hat{f}_{nm}^j\}$, $\{\hat{f}_n^{Ej}\}$, and $\{\hat{A}_n^A\}$, a general equilibrium in relative changes is given by a vector of wages $\{\hat{w}_n^k\}_{n=0,\dots,N}^k$, price indices $\{\hat{P}_{n0}^j\}_{n=0,\dots,N}^j$, $\{\hat{P}_{nC}^j\}_{n=0,\dots,N}^j$, labor allocation $\{\hat{L}_n^k\}_{n=0,\dots,N}^k$, mass of entrants $\{\hat{T}_n^j\}_{n=0,\dots,N}^j$, and cutoff productivity $\{\hat{\phi}_{mn}^j\}_{m,n=0,\dots,N}^j$. See Appendix E for details.

3.6 Decomposition of the Change in Real Wage

Following Caliendo et al. (2023), we solve for changes in the price index and real wage of Chinese region n due to changes in sunk entry costs, trade costs, and employments. Using the conditions (13), (15), (19), and (22), we can obtain the following expression for the expenditure share on domestic inputs:

$$\begin{aligned} \tilde{\lambda}_{nn}^j &= \lambda_{nn}^j g_{nC}^j \\ &= \frac{\theta^j}{\theta^j + 1 - \sigma^j} T_n^j \left(\frac{\tilde{b}_n^j}{\phi_{nn}^j} \right)^{\theta^j} \frac{\sigma^j w_n^M f_{nn}^j}{X_{nC}^j} g_{nC}^j. \end{aligned} \quad (40)$$

After putting condition (19), (15), and (13) again into the (40), we can solve for the change

in price index as

$$\begin{aligned}
\frac{dP_n^j}{P_n^j} = & \underbrace{\frac{1}{\theta^j} \left(\frac{d\tilde{\lambda}_{nn}^j}{\tilde{\lambda}_{nn}^j} - \frac{dT_n^j}{T_n^j} \right)}_{\text{Entry-adjusted domestic share}} - \underbrace{\frac{1}{\theta^j} \left(\frac{\theta^j}{\sigma^j - 1} - 1 \right)}_{\text{Selection}} \underbrace{\left(\frac{dX_n^j}{X_n^j} \right)}_{\text{Scale}} \\
& + \underbrace{\frac{1}{\theta^j} \left(\left(\frac{\theta^j}{\epsilon^j - 1} - 1 \right) - \left(\frac{\theta^j}{\sigma^j - 1} - 1 \right) \right)}_{\text{Armington across regions}} \left(\frac{d\lambda_{nn}^j}{\lambda_{nn}^j} \right) \\
& + \underbrace{\frac{1}{\theta^j} \left(\left(\frac{\theta^j}{\rho^j - 1} - 1 \right) - \left(\frac{\theta^j}{\sigma^j - 1} - 1 \right) \right)}_{\text{Armington across countries}} \left(\frac{dg_{nC}^j}{g_{nC}^j} \right) \\
& + \underbrace{\frac{1}{\theta^j} \left(\frac{\theta^j}{\sigma^j - 1} - 1 \right)}_{\text{Selection}} \underbrace{\left(\frac{d \left(\frac{w_n^M f_{nn}^j}{w_n^M f_{nn}^j} \right)}{\frac{w_n^M f_{nn}^j}{w_n^M f_{nn}^j}} \right)}_{\text{Fixed costs}} + \underbrace{\beta^j \frac{dw_n^M}{w_n^M}}_{\text{Primary costs}} + \underbrace{(1 - \beta^j) \sum_{i \in \{1, 2, \dots, J\}} \gamma^{ji} \frac{dP_n^i}{P_n^i}}_{\text{Intermediate goods costs}}. \quad (41)
\end{aligned}$$

The change in the price index $\frac{dP_n^j}{P_n^j}$ can be decomposed into several structural components. First, the *selection effect*, is embedded across multiple components through coefficients that reflect how firm entry and exit reshape average productivity (Di Giovanni and Levchenko, 2013).² Trade liberalization, market expansion, or reduced fixed costs raise the entry threshold, forcing out less productive firms and lowering prices via increased average efficiency. Conversely, tighter entry barriers or contracting markets lower the cutoffs, reduce competition, and put upward pressure on prices.

On the demand side, changes in the entry-adjusted domestic expenditure share and improvements in domestic technology reduce the price index by increasing the relative competitiveness of local producers. Market expansion, captured by the scale effect, leads to tougher competition and lower markups when the elasticity of substitution across firms, σ^j , exceeds one. Shifts in consumption patterns—reflected in changes in the unadjusted domestic share and trade cost-related parameters—further influence the price index, depending on the gap between

²In the Melitz model, firm productivity heterogeneity is defined by θ^j (higher θ^j means less heterogeneity and thinner tail), while market competition by σ^j (higher σ^j means more intense competition). The strength of the selection effect hinges on the elasticity of firm sales to productivity, given by $\frac{\theta^j}{\sigma^j - 1}$. Higher elasticity allows minor productivity gains yield disproportionate sales increases, leading to a strong selection effect where efficient firms thrive and less efficient ones exit, boosting average productivity. (To illustrate this elasticity more intuitively, imagine a height contest among several people: with σ^j fixed and θ^j very large, everyone's height is almost identical—so if one person grows a little taller, they immediately stand out; similarly, with θ^j fixed and σ^j very small, the judges use very few criteria, so even a slight edge in height makes someone noticeable.) As Di Giovanni and Levchenko (2013) note, when $\theta^j \rightarrow \sigma^j - 1$, this elasticity approaches 1, meaning the selection effect virtually vanishes. Productivity differences no longer significantly drive market share reallocation. In this scenario, the firm size distribution becomes extremely fat-tailed (few giants, many small firms). Trade gains then depend solely on the intensive margin of these very large firms, as they expand existing sales due to broader market access or lower costs, rather than by displacing competitors.

elasticities across firms, origins, and varieties.

On the cost side, rising primary input costs, such as wages and fixed entry costs, exert upward pressure on prices. Similarly, changes in the cost of intermediate inputs feed through the input–output structure and amplify or mitigate price movements depending on sectoral linkages. These channels jointly determine the magnitude and direction of price index changes, and their relative importance depends on underlying structural parameters and the nature of the shocks.

Then we solve the change of total expenditure implied by conditions (24), (29), and (31):

$$dX_n^j = \alpha^j d(HE_n) + \sum_{i=1}^J \frac{\frac{\sigma^i \theta^i}{\sigma^i - 1} (1 - \beta^i) \gamma^{ij}}{\mu^i} (w_n^M f_n^{iE} dT_n^i + T_n^j f_n^{iE} dw_n^M + T_n^i w_n^M df_n^{iE}). \quad (42)$$

Let $\tilde{S}_n^j = \frac{\frac{\sigma^i \theta^i}{\sigma^i - 1} (1 - \beta^i) \gamma^{ij} T_n^i w_n^M f_n^{iE}}{\mu^i X_n^j}$ and $(1 - \tilde{S}_n^j) = \frac{\alpha^j HE_n}{X_n^j}$. Then we rewrite (42):

$$\frac{dX_n^j}{X_n^j} = (1 - \tilde{S}_n^j) \frac{d(HE_n)}{HE_n} + \sum_{i=1}^J \tilde{S}_n^j \left(\frac{dT_n^i}{T_n^i} + \frac{d(w_n^M f_n^{iE})}{w_n^M f_n^{iE}} \right)$$

Finally, we recall the real wage by (3), (4), and (34):

$$v_n^M = \frac{w_n^M}{\prod_j (P_n^j)^{\alpha^j}}. \quad (43)$$

The change in regional manufacturing real wage is obtained by differentiating the Cobb-

Douglas utility function (43) and combining with (41) and (42):

$$\begin{aligned}
\frac{dv_n^M}{v_n^M} &= \frac{dw_n^M}{w_n^M} - \sum_j \alpha^j \frac{dP_n^j}{P_n^j} \\
&= \frac{dw_n^M}{w_n^M} \\
&\quad - \underbrace{\sum_j \frac{\alpha^j}{\theta^j} \left(\frac{d\tilde{\lambda}_{nn}^j}{\tilde{\lambda}_{nn}^j} - \frac{dT_n^j}{T_n^j} \right)}_{\text{Entry-adjusted domestic share}} \\
&\quad + \underbrace{\sum_j \frac{\alpha^j}{\theta^j} \underbrace{\left(\frac{\theta^j}{\sigma^j - 1} - 1 \right)}_{\text{Selection}} \left(\underbrace{\left(1 - \tilde{S}_n^j \right) \frac{d(L_n^M w_n^M + L_n^A w_n^A)}{L_n^M w_n^M + L_n^A w_n^A}}_{\text{Scale}} + \sum_{i=1}^J \tilde{S}_n^j \left(\underbrace{\frac{dT_n^i}{T_n^i}}_{\text{Entry}} + \underbrace{\frac{d(w_n^M f_n^{iE})}{w_n^M f_n^{iE}}}_{\text{Entry costs}} \right) \right)}_{\text{Scale}} \\
&\quad - \sum_j \frac{\alpha^j}{\theta^j} \underbrace{\left(\left(\frac{\theta^j}{\epsilon^j - 1} - 1 \right) - \left(\frac{\theta^j}{\sigma^j - 1} - 1 \right) \right)}_{\text{Armington across regions}} \left(\frac{d\lambda_{nn}^j}{\lambda_{nn}^j} \right) \\
&\quad - \sum_j \frac{\alpha^j}{\theta^j} \underbrace{\left(\left(\frac{\theta^j}{\rho^j - 1} - 1 \right) - \left(\frac{\theta^j}{\sigma^j - 1} - 1 \right) \right)}_{\text{Armington across countries}} \left(\frac{dg_{nC}^j}{g_{nC}^j} \right) \\
&\quad - \sum_j \frac{\alpha^j}{\theta^j} \underbrace{\left(\frac{\theta^j}{\sigma^j - 1} - 1 \right)}_{\text{Selection}} \underbrace{\left(\frac{d(w_n^M f_{nn}^j)}{w_n^M f_{nn}^j} \right)}_{\text{Fixed costs}} \\
&\quad - \sum_j \alpha^j \underbrace{\left(\beta^j \frac{dw_n^M}{w_n^M} \right)}_{\text{Primary costs}} + \underbrace{(1 - \beta^j) \sum_{i \in \{1, 2, \dots, J\}} \gamma^{ji} \frac{dP_n^i}{P_n^i}}_{\text{Intermediate goods costs}}
\end{aligned} \tag{44}$$

In this decomposition, an increase in the domestic consumption share (λ_{nn}^j and g_{nC}^j) can lead to a reduction in real wages, which can be understood through several mechanisms. Firstly, a higher domestic consumption share implies a relatively smaller share of imported goods, potentially diminishing the competitive pressure from firms from other regions or Row. In Melitz models, imports typically introduce new varieties and intensify market competition. Thus, a decline in import share might allow domestic firms to face less competition and potentially command higher markups. Secondly, if consumers allocate a greater portion of their expenditure towards domestically produced goods that may now face reduced competitive discipline, the overall price level of the consumption basket could effectively rise. With nominal

wages held constant, an increase in the consumption price level directly translates to a decrease in real wages.

4 Quantification

We bring the model to the data from the Chinese economy and the RoW. Our analysis covers 29 provinces of Mainland China (listed in Table 2) and six manufacturing sectors (listed in Table 3).³ The sectoral classification follows the input-output tables, which we discuss in detail below.

Table 2: List of the Chinese Provinces

#	Region	#	Region
1	Beijing	16	Henan
2	Tianjin	17	Hubei
3	Hebei	18	Hunan
4	Shanxi	19	Guangdong
5	Inner Mongolia	20	Guangxi
6	Liaoning	21	Chongqing
7	Jilin	22	Sichuan
8	Heilongjiang	23	Guizhou
9	Shanghai	24	Yunnan
10	Jiangsu	25	Yunnan
11	Zhejiang	26	Gansu
12	Anhui	27	Qinghai
13	Fujian	28	Ningxia
14	Jiangxi	29	Xinjiang
15	Shandong		

4.1 Data

We use the *Chinese Industrial Enterprises Database* (CIED) from 2002–2007 to calibrate structural parameters and establish baseline equilibrium outcomes. This dataset covers over 180,000 manufacturing establishments per year in mainland China with sales exceeding 5 million Chinese Yuan (approximately 0.68 million US dollars), a threshold that applies only to privately owned firms. The database provides key establishment information, including location, industry, ownership type, and number of workers, as well as revenue and production costs, which we use to identify structural parameters of the model.

To calibrate sunk entry shocks, we utilize the *Chinese Enterprise Registration Data* (CERD) from 2002–2007, which contains comprehensive registration and deregistration records for all

³Due to data limitations, we have excluded Hainan Island and Tibet from our study.

Table 3: List of Sectors

Abbrev	Sector	MRIO 2002	MRIO 2007	WIOD
Agriculture	Agriculture, Forestry, and Fisheries	1	1	A
Food	Food and Textile	3, 4	6, 7, 8	C10-C15
Chemicals	Chemical and Petroleum	7, 8	11, 12	C19-C20
Metals	Metals and Mineral Processing	9, 10, 11	13, 14, 15	C23-C25
Machinery	Machinery and Equipment	12, 13	16, 17	C28-C30
Electronics	Electrical and Electronic	14, 15, 16	18, 19, 20	C26-C27
Others	Paper, Wood, and Other	5, 6, 17	9, 10, 21	C16-C18, C21-C22, C31-C33

establishments in China.

Additionally, we incorporate data from the *Chinese Multi-Regional Input-Output* (MRIO) tables for 2002 and 2007, along with the *World Input-Output Database* (WIOD), to construct bilateral trade flows across regions and calibrate production function parameters. These bilateral trade flows allow us to compute trade shares and infer iceberg trade costs, as detailed below.

One limitation of the MRIO tables is that they do not provide sector-level import and export data for each Chinese region. To address this, we supplement the input-output tables with the *China Customs Data* from 2002 and 2007, enabling us to construct sector- and region-specific import and export values for Chinese regions.⁴

In addition, we use data from the *2000 Fifth National Population Census* and the *2005 1% National Population Sample Survey* in China to calculate the share of urban and rural populations in each province. Following the approach of [Tombe and Zhu \(2019\)](#), we assume that the population structure in 2000 is representative of the situation in 2002, while the 2005 data serve as a proxy for the population distribution in 2007.

4.2 Calibration

We conduct counterfactual analyses using exact hat algebra, which allows us to compute equilibrium outcomes without directly identifying fundamentals such as fixed costs and amenity shocks. Below, we outline the calibration of structural parameters and baseline equilibrium outcomes, followed by the construction of shocks to fundamentals in the counterfactual analysis.

Structural Parameters

Cobb-Douglas preference parameters are directly observable from the input-output table. We compute the manufacturing sectoral expenditure shares α^j for both China and the RoW using the 2002 WIOD, based on final spending in each sector. We sourced the agricultural expenditure share α^A from [Tombe and Zhu \(2019\)](#). Table 4 summarizes the results.

⁴The dataset includes HS codes for each transaction. We use the concordance table provided by the [WTO website](#).

Table 4: Parameters for α^j

	Sector	α^j
Chinese Regions	Agriculture	0.095
	Food	0.324
	Chemicals	0.018
	Metals	0.038
	Machinery	0.287
	Electronics	0.166
	Others	0.072
RoW	Agriculture	0.095
	Food	0.294
	Chemicals	0.072
	Metals	0.035
	Machinery	0.252
	Electronics	0.128
	Others	0.124

We estimate the elasticity of substitution across varieties σ^j based on firm profit maximization, which implies that revenue is proportional to total variable costs. Using this relationship, we estimate the following regression with CIED data:

$$r_{nlt}^j = \mu^j \times tvc_{nlt}^j + Error_{nlt}^j,$$

where r_{nlt}^j denotes the revenue of firm l in sector j in region n in year t , tvc_{nlt}^j represents total variable costs (sum of total wage payments and intermediate input purchases). Table 5 presents the results.

Table 5: Estimates of σ^j

	μ^j	Standard Error	σ^j
Food	1.217	0.001	5.608
Chemicals	1.226	0.001	5.420
Metals	1.215	0.001	5.648
Machinery	1.199	0.001	6.031
Electronics	1.195	0.001	6.132
Others	1.213	0.001	5.690

We use WIOD data to calibrate labor share β^j and intermediate input shares γ^{ji} . Table 6 summarizes the results. We set labor share $\beta^A = 0.509$ sourced from [Tombe and Zhu \(2019\)](#).⁵

⁵Following [Tombe and Zhu \(2019\)](#), labor's share of output in agriculture is 0.29 and land's share is 0.28. Since we assume that the agricultural sector does not use intermediate inputs, total factor payments only

Table 6: Parameters in Cobb-Douglas Production Function

	Sector	Intermediate Input Shares						
		Labor	Food	Chemicals	Metals	Machinery	Electronics	Others
Chinese Regions	Food	0.279	0.722	0.121	0.029	0.024	0.011	0.093
	Chemicals	0.225	0.030	0.722	0.063	0.046	0.033	0.107
	Metals	0.262	0.011	0.106	0.700	0.066	0.023	0.094
	Machinery	0.274	0.013	0.042	0.322	0.440	0.090	0.094
	Electronics	0.225	0.005	0.056	0.217	0.048	0.554	0.120
	Others	0.329	0.053	0.287	0.066	0.027	0.021	0.545
RoW	Food	0.310	0.658	0.093	0.062	0.022	0.013	0.152
	Chemicals	0.284	0.035	0.730	0.062	0.019	0.016	0.138
	Metals	0.348	0.009	0.076	0.764	0.047	0.033	0.072
	Machinery	0.315	0.015	0.026	0.246	0.521	0.091	0.102
	Electronics	0.345	0.008	0.049	0.182	0.048	0.607	0.105
	Others	0.388	0.042	0.220	0.099	0.041	0.041	0.558

We measure the dispersion of the Pareto distribution using the expression $(\theta^j - \sigma^j + 1)$ following [Helpman et al. \(2004\)](#) and [Bernard et al. \(2018\)](#). We first compute the dispersion measure based on the standard deviation of the logarithm of firm size. Specifically, we infer the dispersion parameter from the slope of a Pareto distribution fitted to the firm size distribution. For each market j , we select all firms with more than 50 employees and rank them by sales. We then estimate the following regression:

$$\log(\text{Rank}_l^j) = \Gamma^j \log(\text{Firm Size}_l^j) + \varepsilon_l^j,$$

where the of the estimated slope yields dispersion. As an additional method, we take the natural logarithm of each firm's sales and calculate the standard deviation within each market j : $\theta^j - \sigma^j + 1 = \text{std}(\log(\text{Firm Size}_l^j))$ by using *CIED* in year of 2002, and the result is very close to the first method.⁶

We then source the parameter of trade elasticity ξ^j from [Bartelme et al. \(2025\)](#) and [Sogalla \(2023\)](#) to back out the upper-tier CES elasticity across countries within each sectors, $\epsilon^j = 1 + \left(\frac{1}{\xi^j \theta^j} - \frac{1}{\theta^j} + \frac{1}{\sigma^j - 1}\right)^{-1}$.⁷

Finally, we calibrate the elasticity of substitution ρ^j between domestic and foreign inputs following [Hillberry and Hummels \(2013\)](#), [Feenstra et al. \(2018\)](#), and [Caliendo et al. \(2023\)](#), who support the "rule of two," where the ratio of ϵ^j to ρ^j is 2.

Regarding the urban and rural choice elasticities, η , we follow the approach of [Doi and](#)

include labor and land. Therefore, the effective labor share in agricultural value added is calculated as $\beta^A = \frac{0.29}{0.29+0.28} \approx 0.509$

⁶As a comparison, we also calibrate the Pareto shape parameter, θ^j , to match $\frac{\sigma^j \theta^j}{\sigma^j - 1} = 5.55$ following ([Chaney, 2008](#); [Eaton et al., 2011](#); [Caliendo et al., 2023](#)).

⁷To introduce scale elasticity and trade elasticity in a consistent manner, we follow the structure outlined in [Kucheryavyy et al. \(2023\)](#) which can be found in Appendix D.

Table 7: Parameters: θ^j , ϵ^j , and ρ^j

	$\theta^j - \sigma^j + 1$	Standard Error	θ^j	ξ^j	ϵ^j	ρ^j
Food	0.837	0.005	5.446	0.615	4.012	2.006
Chemicals	0.798	0.006	5.218	0.615	3.888	1.944
Metals	0.830	0.006	5.478	0.590	3.924	1.962
Machinery	0.802	0.006	5.833	0.607	4.226	2.113
Electronics	0.759	0.007	5.891	0.690	4.688	2.344
Others	0.817	0.006	5.506	0.578	3.889	1.945

Suzuki (2025) and Suzuki (2023), who estimated an elasticity of 0.64 and 0.49, respectively, between Japanese prefectures. For comparison, Balboni (2025) report an elasticity of 0.33 for Vietnam, Fajgelbaum et al. (2019) report an elasticity of 1.8 for U.S. states, while Tombe and Zhu (2019) find an elasticity of 1.5 for eight regions in China, depending on the specification used. However, since we are working with Chinese provinces-area pairs which generally exhibit more heterogeneity and local differences in economic characteristics, the urban and rural choice elasticity is expected to be lower than the values reported for broader regions. Here, we index variables by the time subscript t . From the equilibrium (32), we have

$$\begin{aligned}\psi_{nt}^k &= \frac{B_n^k \left(\frac{w_{nt}^k}{P_{nt}^k} \right)^\eta}{\sum_{m \in \{1, 2, \dots, N\}} B_m^k \left(\frac{w_{mt}^k}{P_{mt}^k} \right)^\eta} \\ &= \frac{B_n^k \left(\frac{w_{nt}^k}{P_{nt}^k} \right)^\eta}{\Phi_t},\end{aligned}$$

where Φ_t is the “multilateral resistance” term. After taking logs, we have the following moment condition:

$$\ln(\psi_{nt}) = \eta \ln \left(\frac{\bar{w}_{nt}}{P_{nt}} \right) + \ln(B_n) + \ln(\Phi_t) + v_{nt},$$

where we consider the unobserved amenity shock $\ln(B_n)$ as the region fixed effects, while the unobserved multilateral resistance term Φ_t as time fixed effects. Any unobserved shocks to the probability of choosing a location may influence average wages. For example, if a positive shock to a region draws more manufacturing workers, the increased labor supply could drive down average manufacturing wages. This would lead to downward bias in the parameter estimates under OLS. On the other hand, unobserved factors such as technological advancements or economic booms could drive up wages, resulting in upward bias in the OLS estimates.

To address potential endogeneity concerns—particularly the reverse causality between

local wages and regional sectoral shares—we construct an instrumental variable that captures exogenous variation in wages. Specifically, we use the log of the national average wage (excluding the focal region) as an instrument for the log real wage, $\bar{w}_{t,-n}^k = \frac{1}{N-1} \sum_{m \neq n} w_{mt}^k$. This Bartik-style instrument isolates wage shocks that originate from broader national trends rather than local unobserved factors, and thus serves to mitigate bias arising from endogenous local labor supply and demand dynamics. By construction, the instrument is strongly correlated with local wages through national labor market integration, while remaining plausibly exogenous to region-specific shocks that directly influence the sectoral composition.

We used CIED to calculate the sectoral wages and area choice probabilities for each region from 2002 to 2007, and we employed the agricultural and industrial price index from the [Wind Financial Terminal](#). The data assumes that the price index for each region was 1 in 1978 (the year China first officially opened up to the world), allowing for comparisons both across regions and over time.

Table 8 presents the estimation results. The OLS estimate of η is lower than the IV estimate, consistent with the previous discussion. The IV estimate is 0.335, which we adopt as the baseline parameter value.

Table 8: Urban and Rural Choice Elasticity (IV)

	OLS	IV
$\ln \left(\frac{w_{nt}^k}{P_{nt}^k} \right)$	0.274	0.335
	(0.119)	(0.133)
Region FE	✓	✓
Year FE	✓	✓
<i>First Stage</i>		
$\bar{w}_{t,-n}^k$		-56.783
		(2.096)
F-statistics		482.28
Observations	240	240

The urban and rural choice elasticities are externally calibrated. We consider two aspects in our calibration. First, if the choice were purely between agriculture and manufacturing sectors, we could follow the approach of [Fan \(2019\)](#), who used Chinese data to regress the log wage of stayers on regional fixed effects, individual characteristics, and industry fixed effects. Second, our focus is on the location choice between urban and rural areas, which, while sharing some features with sectoral mobility, involves distinct considerations such as infrastructure access, public service provision, and amenity differences.

Additionally, we calibrate the change of amenity shock $\hat{B}_n^k$. To recover the amenity shock in each year, we employ a fixed-point iteration method based on the observed location

probabilities ψ_{nt}^k , real wages $\frac{w_{nt}^k}{P_{nt}^k}$, and the Fréchet dispersion parameter η . Starting from an initial guess $B_n^{k,(0)} = 1$ for all regions n , we iteratively update the values according to the following expression:

$$B_n^{k,(s+1)} = \psi_n^k \left(\sum_{m=1}^N B_m^{k,(s)} \left(\frac{w_m^k}{P_m^k} \right)^\eta \right) \left(\frac{P_n^k}{w_n^k} \right)^\eta$$

where the superscript (s) denotes the iteration step. The process continues until convergence. We calibrate this shock to assess whether incorporating it allows our model to accurately capture the observed changes in GDP and the number of firms. However, this shock is excluded from the counterfactual analysis.

Baseline Outcomes

For counterfactual, we set 2002 as the base year and construct baseline equilibrium variables, including the expenditure share on China-sourced goods g_{nC}^j , domestic trade share λ_{nm}^j , worker spending share $\frac{HE_n^j}{X_n^j}$, firm spending share $\frac{ME_n^j}{X_n^j}$, intermediate input spending share $\frac{ME_n^{ij}}{ME_n^j}$, regional sales share $\frac{X_{nm}^j}{Y_n^j}$, and location choice probability ψ_n^k .⁸ We derive these values using MRIO data, supplemented with the Customs Data, CIED, and *National Population Census* data.

Shocks to Fundamentals

We assess how Chinese regional economies would have evolved if manufacturing employments, trade costs, and entry costs experienced the shocks from big reforms.

To do so, we treat the expansion in total population as an exogenous shock in the counterfactual scenario. In our setup, China's total labor L_C is taken as given. We use data *2000 China National Population Census* and *2005 China 1% National Population Sample Survey* to calculate total labor. We follow [Tombe and Zhu \(2019\)](#) in assuming that the labor structure in 2000 (2002) is representative of that in 2005 (2007), and that the relative change in labor, $\frac{L_{C,2005}}{L_{C,2000}} \approx \frac{L_{C,2007}}{L_{C,2002}} = 1.348$, reflects labor growth during this period.

Following [Head and Ries \(2001\)](#), we assume symmetric trade costs ($\tau_{nm}^j = \tau_{mn}^j$) and domestic trade costs normalized to 1 ($\tau_{nn}^j = 1$)⁹ and express bilateral trade costs τ_{nm}^j using

⁸We use *National Population Census* data in 2000 to calculate regional population levels. Assuming the population structure in 2000 is representative of the year 2002, we compute the region-area share as $\psi_{n,2002}^k \approx \psi_{n,2000}^k$.

⁹As a robustness check, we also calibrate trade costs using a gravity model as in [Eaton and Kortum \(2002\)](#), finding a close correlation (0.96) between the two methods.

bilateral trade flows X_{nm}^j :

$$\tau_{nm}^j = \left(\frac{X_{nm}^j X_{mn}^j}{X_{nn}^j X_{mm}^j} \right)^{\frac{1}{2(1-\sigma^j)}}.$$

Figure A4 presents the relative changes in iceberg trade costs, $\hat{\tau}_{mn}^j = \tau_{mn,2007}^j / \tau_{mn,2002}^j$. Values below 1 (blue) indicate declining trade costs over the period, reflecting infrastructure improvements and trade liberalization. The numbers on the axes represent the 29 Chinese provinces and the RoW. Among the results for these six sectors, most regions experienced a reduction in trade costs. However, trade costs in the electronics manufacturing sector have increased.

Finally, we consider a counterfactual scenario where entry costs stay at their post-reform levels. From the equilibrium condition (31), the relative change in the sunk entry cost can be expressed as

$$\hat{f}_m^{jE} = \frac{\hat{Y}_m^j}{\hat{T}_m^j \hat{w}_m} = \frac{\frac{Y_{m,2007}^j}{Y_{m,2002}^j}}{\frac{T_{m,2007}^j}{T_{m,2002}^j} \frac{w_{m,2007}^M}{w_{m,2002}^M}},$$

where $Y_{m,2007}^j$ and $Y_{m,2002}^j$ represent sector j 's output in region m in years 2007 and 2002, respectively. Similarly, $T_{m,2007}^j$ and $T_{m,2002}^j$ denote the mass of entrants in these years, while $w_{m,2007}^M$ and $w_{m,2002}^M$ capture the average wage. We assume that the mass of entrants T_m^j are proportional to the observed number of registered establishments in the dataset and calculate the mass of entrants in each region annually by cumulatively summing the registered establishments year by year and subtracting the number of deregistered establishments for each corresponding year from CERD.¹⁰ We obtain data on total output Y_m^j and from *CIED*. The average wage is recovered by using the IO table implied by equation (36).

Table A3 presents the relative changes in sunk entry costs. We observe that for most regions and sectors, entry costs have decreased. However, in some regions (i.e., Heilongjiang and Guangxi), entry costs have notably increased in chemicals and metals. This is due to

¹⁰According to *CERD*, between 2002 and 2007, approximately 13% of registered enterprises (about 2.31 million) disclosed both the number of insured employees and the range of registered employees. Among these, 33% reported fewer than 10 insured employees, and 22% reported zero insured employees. At the same time, 16% of enterprises indicated that their employee range was at the lowest category, that is, fewer than 50 employees. Additionally, approximately 58% of registered establishments disclosed their registered capital. Among these, 51% reported registered capital of less than 100,000 Chinese Yuan, and 26% reported 10,000 Chinese Yuan or less. We assume that these establishments may be "shell" firms. Given their small number of workers and low registered capital, it is plausible that these firms lack labor and the necessary assets for production, such as equipment and raw materials. These establishments incur registration costs but do not engage in actual production activities. This assumption aligns with the concept of the "mass of entrants" in our framework, where firms enter the market primarily to benefit from registration or establish a legal presence without the intention of substantive operation.

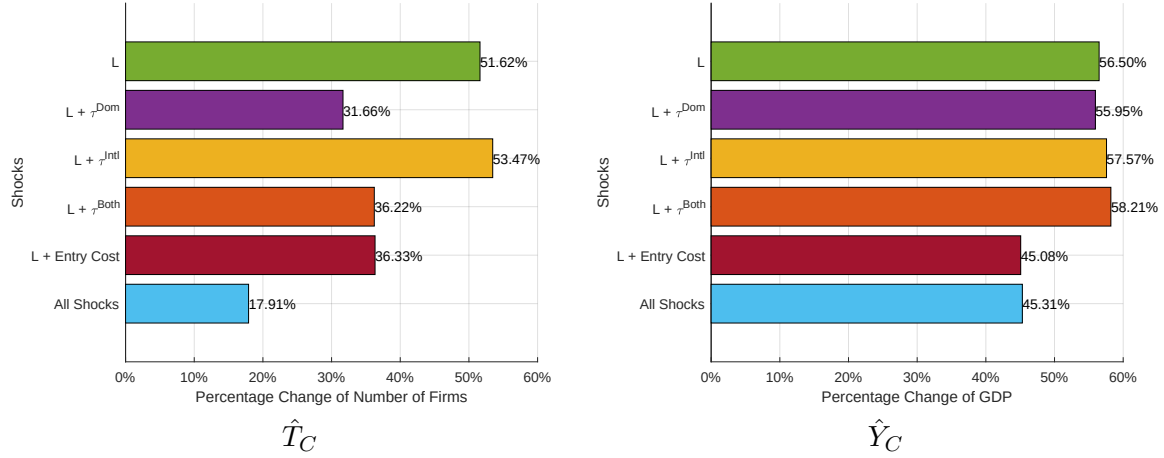
slower GDP growth compared to wage and enterprise growth in these more remote regions. Additionally, entry costs in many regions for the electronics manufacturing sector have increased, which is consistent with the rise in trade costs for this sector that we observed earlier. This could be attributed to higher regional entry barriers, such as stricter regulations, limited access to skilled labor, and higher costs of establishing production facilities, may have contributed. Moreover, the electronics sector might face sector-specific challenges, such as the need for more sophisticated infrastructure, which could increase entry costs in certain regions.

4.3 Counterfactuals

Nationwide Counterfactual Outcomes

Using 2002 as the base year, we conduct counterfactual analyses to assess the role of trade costs, sunk entry costs, and employment in shaping the spatial distribution of economic activity during China's economic reforms. For example, by applying the shock $\hat{\tau}_{mn}^j = \frac{\tau_{mn,2007}^j}{\tau_{mn,2002}^j}$ for $m, n \neq 0$, we simulate how regional economies would have looked if domestic trade costs had remained at their 2007 levels while keeping all other factors unchanged at 2002 levels.

Figure 2: Changes in Number of Firms and GDP in Counterfactuals



The results of changes in number of active firms at the nation-level are shown on the left in Figure 2. When we introduce shocks to employment, trade costs, and sunk entry costs simultaneously, the counterfactual results show a firm growth of 14.66%.

While the green bar indicates a strong 53.64% increase in the number of firms, this growth is contingent upon a sufficient expansion of the labor supply to facilitate more firms to enter the market.

However, when we introduce shocks to domestic trade costs, lowering trade costs increases

competition among firms and the number of firms increased by 35.48%. This aligns with the core idea of Melitz (2003): more open regions face fiercer competition, where the competition from imports is larger than the expansion of export opportunities, pushing out low-productivity firms while allowing productive ones to survive and less productive ones quit. When we introduce shocks to international trade costs, the number of firms increased by 55.49% which is close to the result after only giving the shock to employment. Compared to domestic trade, the competitive pressure brought by international trade is relatively smaller, as domestic trade costs play a more significant role in shaping economy activities (Tombe and Zhu, 2019). At this point, the competition from imports is smaller than the expansion of export opportunities.

When we introduce a sunk entry cost shock, the number of firms increases by 33.23%. This is because lower entry costs reduce the barriers for new firms to enter the market, promoting greater market participation and contributing to an increase in firm numbers. The overall pattern highlights how different types of shocks affect market entry and competition in distinct ways, with expansion of manufacturing employment and reduction of entry cost reductions encouraging firm growth, while competition from domestic trade tends to consolidate the market around more productive firms.

Turning to the right panel of Figure 2, we show the changes in GDP. The impacts of different shocks on GDP are highly similar to the results of different shocks on number of firms. Expansion of manufacturing employment remains the primary driver. Lower trade costs combine the effects of competition from imports and the expansion of export opportunities. Upon examination, we observe that the GDP growth resulting from the reduced sunk entry cost shock (reaching 45.08%) is remarkably close to the effect of implementing all shocks combined. This strong similarity underscores the substantial impact of lowered entry barriers in driving economic expansion. The magnitude of this effect highlights that facilitating firm entry, by reducing sunk costs, plays a pivotal role, nearly as influential as the entire set of considered economic stimuli.

When sunk entry costs fall, the effects on GDP are more nuanced. Lower entry costs encourage more firms to enter the market, intensifying competition. While this increase in firm numbers can drive innovation and efficiency, the heightened competition may also lead to mixed outcomes—some firms thrive and contribute positively to GDP, while others may struggle to survive in a more competitive environment. Overall, the net effect on GDP depends on how well firms adapt to these changing market conditions.

Regional and Sectoral Counterfactual Outcomes

We focus on two representative provinces: Liaoning (northeastern coastal) and Sichuan (southwestern inland). We report the percentage changes from the base year in the number of firms $\hat{T}_{nn}^j$. To support our conclusions, we also examine changes in manufacturing employment

$\hat{L}_n^M$, sectoral GDP $\hat{Y}_n^j$, and the average productivity $\hat{\phi}_{nn}^j$. The results are shown in Tables 9 and 10.

Overall, after incorporating all shocks—shocks to manufacturing employment, trade costs, and entry costs—the changes in the number of firms predicted by the model closely align with those observed in the data. In Liaoning, the number of firms increases by 13.87%, while in Sichuan, the model predicts a 33.3% increase. Looking more closely at the sector level, we find that the predicted changes in the number of firms across sectors after giving all shocks in Liaoning reveal heterogeneous effects: while the number of firms in Liaoning’s food and electronics industries is predicted to decrease, other sectors are expected to see an increase.

The differential impact of each shock closely aligns with the overall national-level results. In both Liaoning and Sichuan, we find that an increase in manufacturing employment is the primary driver of firm growth. In contrast, a reduction in trade costs leads to a decline in the number of firms, as it combines the effects of increased competition from imports and the expansion of export opportunities. Lowering entry costs, however, results in a significant increase in the number of firms.

Next, we highlight the differences between coastal and inland regions. When domestic trade costs decrease, the decline in the number of firms is more pronounced in Sichuan than in Liaoning (Liaoning: 37.74% $\rightarrow$ 8.48% < Sichuan: 42.41% $\rightarrow$ 25.15%). As an inland province, Sichuan’s firms rely more on the local market compared to those in Liaoning. Consequently, the reduction in local trade costs intensifies market competition in Sichuan, leading to the exit of less efficient firms and a larger overall decline in firm numbers.

However, when we lower international trade costs, Sichuan again experiences a greater decline in the number of firms (Liaoning: 37.74% $\rightarrow$ 36.74% < Sichuan: 42.41% $\rightarrow$ 39.62%). This may seem counterintuitive, given that Liaoning, as a coastal province, is more engaged in international trade. While lower international trade costs increase export opportunities, they also heighten competition from foreign firms. Nevertheless, firms in Liaoning are generally more competitive in global markets and thus better equipped to adapt to these changes, resulting in a smaller decline in firm numbers compared to Sichuan. This pattern is further supported by changes in regional GDP. Since firms in coastal regions tend to be more productive, the reduction in both domestic and international trade costs leads to a greater decline in Sichuan’s GDP compared to Liaoning’s.

Furthermore, we find that the model captures each region’s development of its core sectors. If all shocks are applied, the metals sector and chemicals sector in Liaoning would experience the largest increase in the number of firms, rising by 45.43% and 53.20%. This is because metals is a key pillar industry in Liaoning, and as a coastal region, it provides new metals and chemical firms with easier access to international trade.

Meanwhile, in Sichuan, while the metals and chemicals sectors are core industries, the electronics manufacturing sector is experiencing rapid growth, particularly in contrast to

Table 9: Liaoning (Northeastern Coastal)

Sector	Shocks					
	ALL	L	$L + \tau^{Dom}$	$L + \tau^{Int}$	$L + \tau^{Both}$	$L + f^E$
	Manufacturing Employment $\hat{L}_n^M$					
Total	34.08%	35.00%	32.69%	34.37%	32.37%	36.39%
	Number of Firms $\hat{T}_{nn}^j$					
Food	-13.81%	38.23%	-21.64%	28.00%	-28.14%	40.93%
Chemicals	53.20%	36.51%	62.84%	32.40%	62.52%	32.18%
Metals	45.43%	37.05%	35.94%	34.25%	34.83%	50.92%
Machinery	26.05%	39.96%	-6.96%	33.19%	-8.05%	53.17%
Electronics	-72.84%	40.00%	-49.81%	77.26%	-36.60%	-1.40%
Others	-11.29%	33.38%	-1.70%	31.96%	-1.57%	26.44%
Total	13.87%	37.74%	8.48%	36.74%	8.04%	38.54%
	Sectoral GDP $\hat{Y}_n^j$					
Food	-0.86%	43.02%	-8.41%	26.70%	-17.93%	49.43%
Chemicals	136.03%	41.44%	163.65%	33.55%	161.16%	25.80%
Metals	129.65%	43.35%	144.38%	35.48%	135.99%	58.17%
Machinery	72.68%	46.85%	35.83%	36.64%	30.63%	57.47%
Electronics	-66.92%	39.11%	-52.08%	111.39%	-32.45%	-9.66%
Others	1.61%	38.32%	15.34%	41.93%	17.20%	22.60%
Total	73.62%	42.98%	76.41%	41.23%	72.85%	42.10%
	Average Productivity $\hat{\phi}_{nn}^j$					
Food	20.12%	-0.40%	-1.96%	-1.31%	-2.12%	22.89%
Chemicals	16.93%	-0.39%	4.26%	-1.01%	4.41%	10.90%
Metals	21.88%	-0.20%	6.06%	-0.95%	5.83%	17.41%
Machinery	14.70%	-0.14%	1.98%	-0.61%	1.76%	13.11%
Electronics	10.24%	-1.05%	-5.13%	1.97%	-3.11%	8.77%
Others	11.32%	-0.36%	-1.87%	0.21%	-1.35%	12.10%

Note: The table shows changes in outcomes from 2002 to 2007. Number of active firm numbers ($\hat{T}_{nn}^j$), GDP ($\hat{Y}_n^j$), productivity cutoff ($\hat{\phi}_{nn}^j$), and manufacturing employment ($\hat{L}_n^M$) are reported. The following table is the same.

Table 10: Sichuan (Southwestern Inland)

Sector	Shocks					
	ALL	L	$L + \tau^{Dom}$	$L + \tau^{Int}$	$L + \tau^{Both}$	$L + f^E$
	Manufacturing Employment $\hat{L}_n^M$					
Total	39.76%	41.04%	34.19%	38.32%	31.54%	49.06%
	Number of Firms $\hat{T}_{nn}^j$					
Food	13.47%	41.66%	8.03%	40.20%	7.71%	45.36%
Chemicals	48.26%	42.45%	32.10%	40.30%	29.21%	49.81%
Metals	44.27%	41.55%	30.96%	38.11%	29.00%	56.87%
Machinery	16.15%	43.22%	21.04%	41.03%	18.88%	45.58%
Electronics	31.26%	48.59%	49.89%	40.09%	38.16%	54.67%
Others	41.11%	40.27%	18.94%	38.71%	16.86%	59.75%
Total	33.30%	42.41%	25.15%	39.62%	22.53%	51.53%
	Sectoral GDP $\hat{Y}_n^j$					
Food	44.08%	55.64%	21.59%	48.90%	18.11%	83.73%
Chemicals	102.29%	57.50%	51.02%	47.23%	40.76%	100.05%
Metals	93.91%	56.01%	42.27%	45.37%	35.88%	115.68%
Machinery	46.56%	57.56%	43.18%	45.05%	32.48%	85.78%
Electronics	43.79%	59.57%	80.99%	35.04%	45.47%	76.88%
Others	71.86%	55.77%	23.54%	47.51%	16.51%	111.89%
Total	69.47%	56.65%	40.62%	45.65%	30.94%	97.21%
	Average Productivity $\hat{\phi}_{nn}^j$					
Food	18.38%	-0.21%	1.15%	0.18%	1.68%	16.56%
Chemicals	21.07%	-0.10%	1.51%	-0.04%	1.63%	18.74%
Metals	26.75%	-0.15%	0.49%	0.02%	0.93%	25.44%
Machinery	8.06%	-0.17%	1.94%	-0.38%	1.85%	6.79%
Electronics	9.55%	-0.58%	2.28%	-1.47%	0.86%	8.87%
Others	30.80%	-0.01%	-0.32%	0.20%	-0.08%	30.93%

$$\text{Upstream } \tilde{\text{Distance}}_m^j = \sum_i \sum_n \text{Distance}_{nm} \frac{X_{nm}^{ij}}{\sum_i \sum_n X_{nm}^{ij}},$$

where Distance_{nm} represents the geographical distance between location n and location m , while X_{nm}^{ij} denotes the flow of intermediate goods from sector i in location n to sector j in location m from *MRIO*. A smaller weighted distance suggests a stronger tendency for co-location driven by supply chain interdependencies.

Figure 4: Intermediate Input Linkage

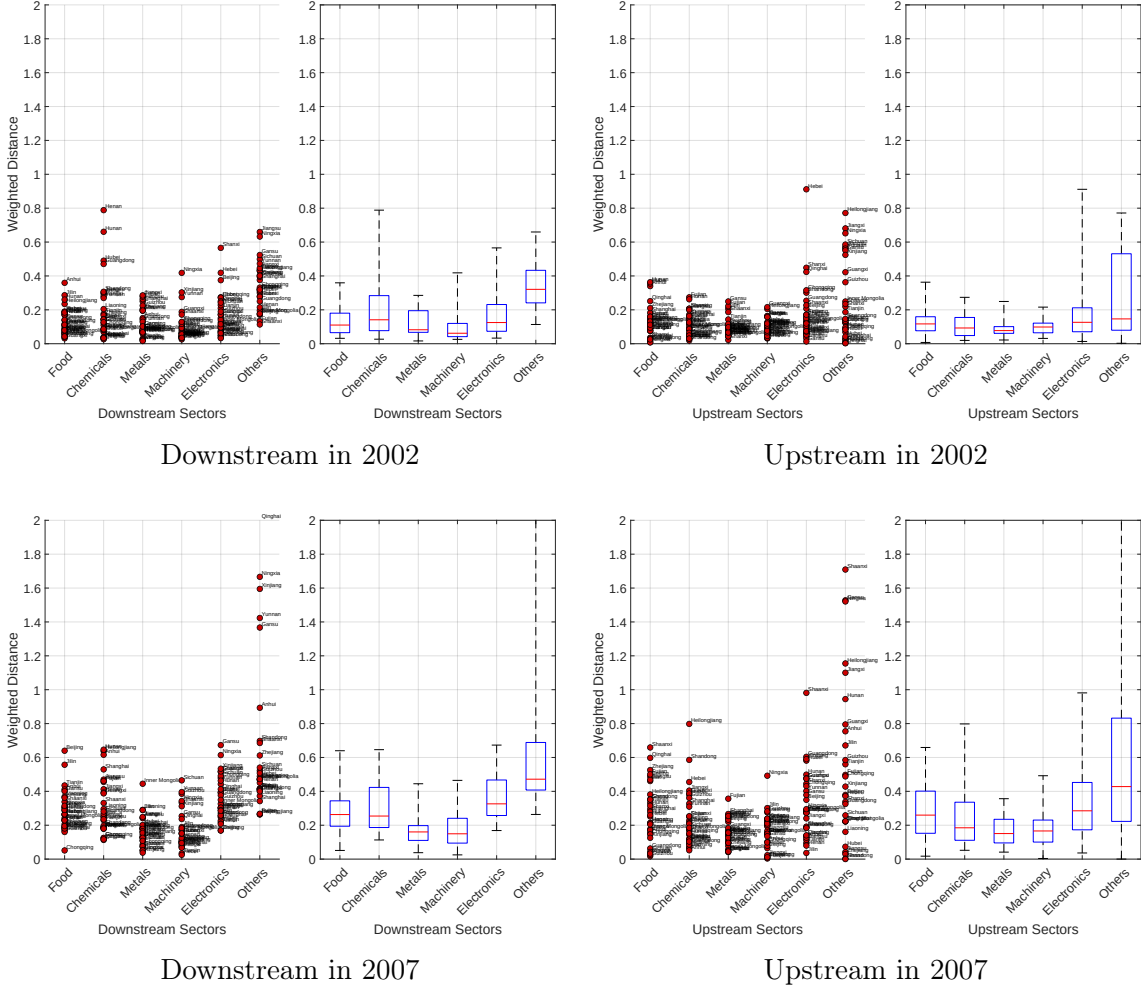


Figure 4 provides compelling evidence for input-output linkage within China's manufacturing landscape. A salient example is observed by the machinery and metals sectors. Compared to other sectors, these two sectors have the shortest average weighted distance when acting as upstream and downstream. And the average weighted distance between the nation's sectors has shortened over time. This finding implies that beyond the inherent self-agglomeration of

individual sectors, there exists a significant inter-sector spatial correlation, wherein sectors with frequent intermediate good transactions tend to be geographically proximate. This is consistent with our finding that industry concentration in the metals and chemical sectors has increased more slowly compared to other sectors.

Welfare

We measure drivers of the change of welfare, as shown in Figure 5 reveals how each shock contributes to the overall welfare change. The baseline scenario with all shocks included shows a welfare increase of 127.09%. When the domestic trade cost shock is removed, welfare drops to 100.94%, indicating a 26.15% reduction. This significant impact shows reduced domestic trade costs enhance competition, driving less productive firms out of the market and allowing more productive firms to thrive.

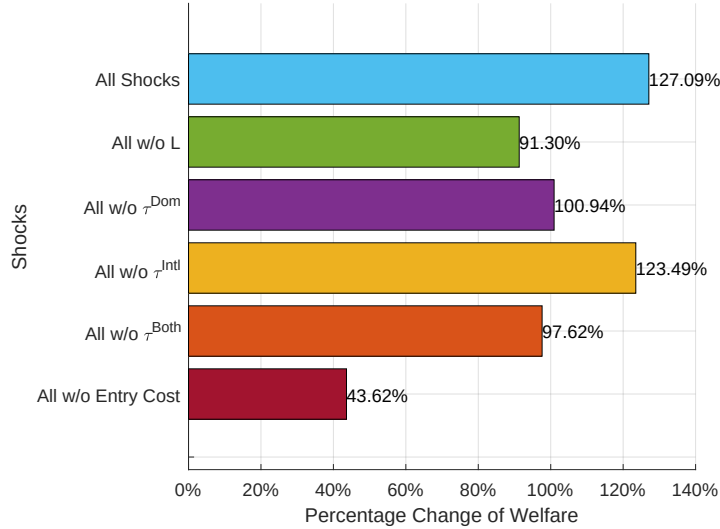


Figure 5: Drivers of the Change of Welfare

In contrast, eliminating the international trade cost shock only reduces welfare to 123.49%. This suggests that while international competition and expanded export opportunities are beneficial, domestic trade liberalization plays a much larger role in boosting welfare.

Excluding the labor supply shock results in a welfare increase of 91.3%, corresponding to an 35.79% reduction relative to the baseline.

The largest impact comes from the sunk entry cost shock. When it is excluded, welfare increases by 43.62%, translating to a 83.47% drop. Lower entry costs make it easier for new firms to enter the market, increasing competition and contributing to welfare improvements.

4.4 Decomposition and Selection

Given the significance of selection effects and the substantial welfare gains observed relative to prior literature, it is essential to identify the sources of these gains. To this end, we decompose the welfare improvements according to the six principal components delineated in Equation (44). For clarity, we refer to these components sequentially: Term 1 corresponds to the first line; Terms 2 and 3 pertain to the second line; and Terms 4 through 6 are detailed in the subsequent lines, as specified in the note accompanying Table 11.

$$\begin{aligned}
\text{Term 1} &= \frac{dw_n^M}{w_n^M} \\
\text{Term 2} &= - \sum_j \frac{\alpha^j}{\theta^j} \underbrace{\left(\frac{d\tilde{\lambda}_{nn}^j}{\tilde{\lambda}_{nn}^j} - \frac{dT_n^j}{T_n^j} \right)}_{\text{Entry-adjusted domestic share}} \\
\text{Term 3} &= \sum_j \frac{\alpha^j}{\theta^j} \underbrace{\left(\frac{\theta^j}{\sigma^j - 1} - 1 \right)}_{\text{Selection}} \underbrace{\left(\left(1 - \tilde{S}_n^j \right) \frac{d(L_n^M w_n^M + L_n^A w_n^A)}{L_n^M w_n^M + L_n^A w_n^A} + \sum_{i=1}^J \tilde{S}_n^j \left(\underbrace{\frac{dT_n^i}{T_n^i}}_{\text{Entry}} + \underbrace{\frac{d(w_n^M f_n^{iE})}{w_n^M f_n^{iE}}}_{\text{Entry costs}} \right) \right)}_{\text{Scale}} \\
&\quad - \sum_j \frac{\alpha^j}{\theta^j} \underbrace{\left(\frac{\theta^j}{\sigma^j - 1} - 1 \right)}_{\text{Selection}} \underbrace{\left(\frac{d(w_n^M f_{nn}^j)}{w_n^M f_{nn}^j} \right)}_{\text{Fixed costs}} \\
\text{Term 4} &= - \sum_j \frac{\alpha^j}{\theta^j} \underbrace{\left(\left(\frac{\theta^j}{\epsilon^j - 1} - 1 \right) - \left(\frac{\theta^j}{\sigma^j - 1} - 1 \right) \right)}_{\text{Armington across regions}} \left(\frac{d\lambda_{nn}^j}{\lambda_{nn}^j} \right) \\
&\quad - \sum_j \frac{\alpha^j}{\theta^j} \underbrace{\left(\left(\frac{\theta^j}{\rho^j - 1} - 1 \right) - \left(\frac{\theta^j}{\sigma^j - 1} - 1 \right) \right)}_{\text{Armington across regions}} \left(\frac{dg_{nC}^j}{g_{nC}^j} \right) \\
\text{Term 5} &= - \sum_j \alpha^j \left(\underbrace{\beta^j \frac{dw_n^M}{w_n^M}}_{\text{Primary costs}} + \underbrace{(1 - \beta^j) \sum_{i \in \{1, 2, \dots, J\}} \gamma^{ji} \frac{dP_n^i}{P_n^i}}_{\text{Intermediate goods costs}} \right)
\end{aligned}$$

Based on the results, selection effects (Term 3) are the dominant contributor to utility improvements, comprising a substantial 67.2% share of the total gains. In contrast, the aggregate contribution of “generic” effects (Terms 1+2+4+5) is notably lower at 0.328. A closer examination of the “generic” components indicates that Term 4 contributes the most, at

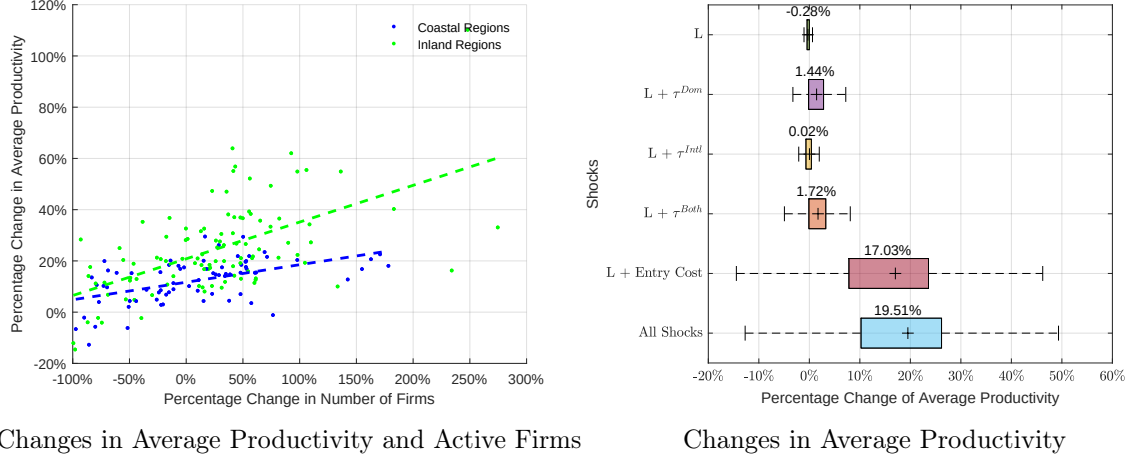
Table 11: Real Wage Decomposition via Contributions of Terms

	Term 1	Term 2	Term 3	Term 4	Term 5	Share of Gains “Generic” Terms w/o 3	“Selection” Term 3
Shock: All							
Real wage	0.028	0.007	0.672	0.317	-0.023	0.328	0.672
Std. Err.	0.022	0.008	0.170	0.775	0.026	0.511	0.170
Observations	29	29	29	29	29	29	29
R^2	0.786	0.654	0.974	0.284	0.657	0.956	0.974
Shock: L							
Real wage	-0.080	-0.188	-0.608	1.867	0.009	1.608	-0.608
Std. Err.	0.014	0.060	0.055	0.869	0.003	0.285	0.055
Observations	29	29	29	29	29	29	29
R^2	0.921	0.773	0.977	0.614	0.829	0.986	0.977
Shock: $L + \tau^{Dom}$							
Real wage	0.042	-0.141	0.737	0.376	-0.014	0.263	0.737
Std. Err.	0.030	0.389	0.231	0.341	0.008	0.116	0.231
Observations	29	29	29	29	29	29	29
R^2	0.745	0.163	0.938	0.644	0.796	0.943	0.938
Shock: $L + \tau^{Int}$							
Real wage	-0.137	-0.236	-0.854	2.211	0.016	1.854	-0.854
Std. Err.	0.017	0.029	0.081	0.872	0.002	0.215	0.081
Observations	29	29	29	29	29	29	29
R^2	0.802	0.797	0.868	0.275	0.838	0.979	0.868
Shock: $L + \tau^{Both}$							
Real wage	0.047	-0.100	0.666	0.402	-0.015	0.334	0.666
Std. Err.	0.029	0.391	0.293	0.280	0.007	0.185	0.293
Observations	29	29	29	29	29	29	29
R^2	0.692	0.053	0.815	0.636	0.813	0.932	0.815
Shock: $L + f^E$							
Real wage	0.065	0.013	1.393	-0.428	-0.043	-0.393	1.393
Std. Err.	0.012	0.003	0.053	0.921	0.011	0.677	0.053
Observations	29	29	29	29	29	29	29
R^2	0.741	0.618	0.986	0.021	0.612	0.946	0.986

Note: Following [Caliendo et al. \(2023\)](#), the 5 coefficients are from a regression of each term on real wage without constant. By construction, the coefficients add up to 100% of the change in real wage. To simplify the results, we assume that $\rho^j = \epsilon^j$. In order to compare the outcomes of different methods, we include all terms simultaneously in a single regression and assess each term’s individual contribution to the model’s overall R^2 . See the Appendix [A5](#) for details.

0.317, thereby highlighting the significant role of Armington effects in driving these “generic” utility improvements. Terms 1 and 2 contribute relatively little (0.028 and 0.007, respectively), and Term 5 even exhibits a negative contribution (-0.023). Overall, these findings underscore the paramount importance of selection effects in explaining welfare gains, with Armington effects being the primary driver within the “generic” category.

Figure 6: Selection Effect (All Shocks)



In the context of rural-urban migration and the reduction of local trade and sunk entry costs, Figure 6 provides critical insights into the differential impact of selection effects across regions. The left panel, illustrating average productivity against firm numbers, reveals a more nuanced story regarding regional disparities. Specifically, the steeper slope observed for inland regions compared to coastal regions signifies a more potent selection effect in the former. This implies that reducing local trade and sunk entry costs in inland areas disproportionately enhances average productivity. Such cost reductions likely enable a more efficient sorting mechanism, allowing a greater share of highly productive firms to emerge and thrive, or conversely, encouraging less efficient firms to exit. And the right panel shows that lower entry costs play a dominant role in raising average productivity.

5 Conclusion

This study investigates how reductions in trade costs, improvements in the business climate, and the expansion of manufacturing employment collectively shaped the geographic distribution of economic activity across Chinese provinces during the country’s economic takeoff in the early 2000s. We develop a spatial general equilibrium model, rooted in the Melitz-Chaney framework, extending it to a multi-sector, multi-region setting and incorporating detailed

sectoral and regional linkages. By calibrating the model to Chinese data and conducting a series of counterfactual simulations, we aim to uncover the mechanisms through which these key policy reforms influenced firm spatial distribution, growth of productivity, regional specialization, and overall welfare.

Our counterfactual analysis reveals the multifaceted impacts on regional development. We find that while reductions in trade costs intensify competition, acting as a deterrent to firm entry. Conversely, significant improvements in the local business climate effectively lower entry barriers, substantially fostering the creation of new firms. Under the combined influence of manufacturing employment expansion, lower trade costs, and an improved business climate, we observe substantial growth in firm numbers and economic activity, particularly in inland regions. In contrast, coastal regions, despite benefiting from expanded international trade opportunities, experience relatively slower firm growth due driven by heightened competition.

Further analysis demonstrates that reductions in entry costs contribute significantly to GDP growth. We also quantify the contributions of manufacturing employment expansion, trade cost reductions, and business climate improvements to overall welfare gains. Notably, our decomposition of real wage growth reveals that selection effects dominate the growth of regional real wages, with these effects being more pronounced in inland regions compared to coastal regions. Collectively, our findings underscore the central role of entry costs, trade costs, and manufacturing employment in shaping the spatial distribution of Chinese manufacturing and driving welfare gains.

References

- Allen, T. and Arkolakis, C. (2014). Trade and the topography of the spatial economy. *Quarterly Journal of Economics*, 129(3):1085–1140.
- Armington, P. S. (1969). A theory of demand for products distinguished by place of production. *Staff Papers-International Monetary Fund*, pages 159–178.
- Au, C.-C. and Henderson, J. V. (2006). How migration restrictions limit agglomeration and productivity in china. *Journal of Development Economics*, 80(2):350–388.
- Balboni, C. (2025). In harm’s way? infrastructure investments and the persistence of coastal cities. *American Economic Review*, 115(1):77–116.
- Baldwin, R. E. and Okubo, T. (2006). Heterogeneous firms, agglomeration and economic geography: Spatial selection and sorting. *Journal of Economic Geography*, 6(3):323–346.
- Bartelme, D., Costinot, A., Donaldson, D., and Rodriguez-Clare, A. (2025). The textbook case for industrial policy: Theory meets data. *Journal of Political Economy*, 133(5):000–000.

- Baum-Snow, N., Brandt, L., Henderson, J. V., Turner, M. A., and Zhang, Q. (2017). Roads, railroads, and decentralization of chinese cities. *Review of Economics and Statistics*, 99(3):435–448.
- Bernard, A. B., Moxnes, A., and Ulltveit-Moe, K. H. (2018). Two-sided heterogeneity and trade. *Review of Economics and Statistics*, 100(3):424–439.
- Brandt, L., Kambourov, G., and Storesletten, K. (2025). Barriers to entry and regional economic growth in china. *Review of Economic Studies*.
- Brandt, L., Van Biesebroeck, J., Wang, L., and Zhang, Y. (2017). Wto accession and performance of chinese manufacturing firms. *American Economic Review*, 107(9):2784–2820.
- Caliendo, L., Feenstra, R. C., Romalis, J., and Taylor, A. M. (2023). Tariff reductions, heterogeneous firms, and welfare: Theory and evidence for 1990–2010. *IMF Economic Review*, 71(4):817–851.
- Caliendo, L., Opromolla, L. D., Parro, F., and Sforza, A. (2021). Goods and factor market integration: A quantitative assessment of the eu enlargement. *Journal of Political Economy*, 129(12):3491–3545.
- Caliendo, L. and Parro, F. (2014). Estimates of the trade and welfare effects of nafta. *Review of Economic Studies*, 82(1):1–44.
- Chaney, T. (2008). Distorted gravity: The intensive and extensive margins of international trade. *American Economic Review*, 98(4):1707–1721.
- Che, Z., Che, J., and Wang, S. (2024). The gains from changes in internal trade costs: A quantitative analysis of china. *Journal of Regional Science*.
- Dekle, R., Eaton, J., and Kortum, S. (2008). Global rebalancing with gravity: Measuring the burden of adjustment. *IMF Staff Papers*, 55(3):511–540.
- Desmet, K., Nagy, D. K., and Rossi-Hansberg, E. (2018). The geography of development. *Journal of Political Economy*, 126(3):903–983.
- Di Giovanni, J. and Levchenko, A. A. (2013). Firm entry, trade, and welfare in zipf’s world. *Journal of International Economics*, 89(2):283–296.
- Doi, Y. and Suzuki, K. (2025). Gains from foreign employment in japan: Regional and sectoral implications. *Working Paper*.
- Donaldson, D. (2018). Railroads of the raj: Estimating the impact of transportation infrastructure. *American economic review*, 108(4-5):899–934.

- Donaldson, D. and Hornbeck, R. (2016). Railroads and american economic growth: A “market access” approach. *Quarterly Journal of Economics*, 131(2):799–858.
- Eaton, J. and Kortum, S. (2002). Technology, geography, and trade. *Econometrica*, 70(5):1741–1779.
- Eaton, J., Kortum, S., and Kramarz, F. (2011). An anatomy of international trade: Evidence from french firms. *Econometrica*, 79(5):1453–1498.
- Ellison, G. and Glaeser, E. L. (1997). Geographic concentration in us manufacturing industries: A dartboard approach. *Journal of Political Economy*, 105(5):889–927.
- Erten, B. and Leight, J. (2021). Exporting out of agriculture: The impact of wto accession on structural transformation in china. *Review of Economics and Statistics*, 103(2):364–380.
- Faber, B. (2014). Trade integration, market size, and industrialization: Evidence from china’s national trunk highway system. *Review of Economic Studies*, 81(3):1046–1070.
- Fajgelbaum, P. D., Morales, E., Suárez Serrato, J. C., and Zidar, O. (2019). State taxes and spatial misallocation. *Review of Economic Studies*, 86(1):333–376.
- Fan, J. (2019). Internal geography, labor mobility, and the distributional impacts of trade. *American Economic Journal: Macroeconomics*, 11(3):252–288.
- Fang, M. and Huang, Z. (2022). Migration, housing constraints, and inequality: A quantitative analysis of china. *Labour Economics*, 78:102200.
- Feenstra, R. C., Luck, P., Obstfeld, M., and Russ, K. N. (2018). In search of the armington elasticity. *Review of Economics and Statistics*, 100(1):135–150.
- Fontagné, L., Guimbard, H., and Orefice, G. (2022). Tariff-based product-level trade elasticities. *Journal of International Economics*, 137:103593.
- Fujita, M., Krugman, P. R., and Venables, A. (2001). *The Spatial Economy: Cities, Regions, and International Trade*. MIT Press.
- Glaeser, E. L., Kallal, H. D., Scheinkman, J. A., and Shleifer, A. (1992). Growth in cities. *Journal of Political Economy*, 100(6):1126–1152.
- Head, K. and Ries, J. (2001). Increasing returns versus national product differentiation as an explanation for the pattern of us–canada trade. *American Economic Review*, 91(4):858–876.
- Helpman, E., Melitz, M. J., and Yeaple, S. R. (2004). Export versus fdi with heterogeneous firms. *American Economic Review*, 94(1):300–316.

- Henderson, J. V., Squires, T., Storeygard, A., and Weil, D. (2018). The global distribution of economic activity: Nature, history, and the role of trade. *Quarterly Journal of Economics*, 133(1):357–406.
- Hillberry, R. and Hummels, D. (2013). Trade elasticity parameters for a computable general equilibrium model. In Dixon, P. B. and Jorgenson, D., editors, *Handbook of Computable General Equilibrium Modeling*, pages 1213–1269. Elsevier.
- Imbert, C., Seror, M., Zhang, Y., and Zylberberg, Y. (2022). Migrants and firms: Evidence from china. *American Economic Review*, 112(6):1885–1914.
- Imbs, J. and Wacziarg, R. (2003). Stages of diversification. *American Economic Review*, 93(1):63–86.
- Kornai, J., Maskin, E., and Roland, G. (2003). Understanding the soft budget constraint. *Journal of Economic Literature*, 41(4):1095–1136.
- Kucheryavyy, K., Lyn, G., and Rodríguez-Clare, A. (2023). Grounded by gravity: A well-behaved trade model with industry-level economies of scale. *American Economic Journal: Macroeconomics*, 15(2):372–412.
- Li, H. and Zhou, L.-A. (2005). Political turnover and economic performance: the incentive role of personnel control in china. *Journal of Public Economics*, 89(9-10):1743–1762.
- Li, X., Ma, L., and Tang, Y. (2024). Migration and resource misallocation in china. *Journal of Development Economics*, 167:103218.
- Lin, Y. (2017). Travel costs and urban specialization patterns: Evidence from china’s high speed railway system. *Journal of Urban Economics*, 98:98–123.
- Liu, C. and Ma, X. (2023). Migration, tariffs, and china’s export surge. *Journal of International Economics*, 140:103696.
- Lu, Y. and Yu, L. (2015). Trade liberalization and markup dispersion: Evidence from china’s wto accession. *American Economic Journal: Applied Economics*, 7(4):221–253.
- Ma, L. and Tang, Y. (2020). Geography, trade, and internal migration in china. *Journal of Urban Economics*, 115:103181.
- Ma, L. and Tang, Y. (2024). The distributional impacts of transportation networks in china. *Journal of International Economics*, 148:103873.
- Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica*, 71(6):1695–1725.

- Montinola, G., Qian, Y., and Weingast, B. R. (1995). Federalism, chinese style: the political basis for economic success in china. *World Politics*, 48(1):50–81.
- Nagy, D. K. (2022). Trade and urbanization: Evidence from hungary. *American Economic Journal: Microeconomics*, 14(3):733–790.
- Perla, J., Tonetti, C., and Waugh, M. E. (2021). Equilibrium technology diffusion, trade, and growth. *American Economic Review*, 111(1):73–128.
- Qian, Y. and Roland, G. (1998). Federalism and the soft budget constraint. *American Economic Review*, pages 1143–1162.
- Qin, Y. (2017). “no county left behind?” the distributional impact of high-speed rail upgrades in china. *Journal of Economic Geography*, 17(3):489–520.
- Redding, S. J. (2016). Goods trade, factor mobility and welfare. *Journal of International Economics*, 101:148–167.
- Redding, S. J. and Rossi-Hansberg, E. (2017). Quantitative spatial economics. *Annual Review of Economics*, 9(1):21–58.
- Redding, S. J. and Sturm, D. M. (2008). The costs of remoteness: Evidence from german division and reunification. *American Economic Review*, 98(5):1766–1797.
- Redding, S. J. and Turner, M. A. (2015). Transportation costs and the spatial organization of economic activity. *Handbook of regional and urban economics*, 5:1339–1398.
- Sogalla, R. (2023). Unilateral carbon pricing and heterogeneous firms. *DIW Berlin Discussion Paper*.
- Suzuki, Y. (2023). Local shocks and regional dynamics in an aging economy. *Working Paper*.
- Tombe, T. and Zhu, X. (2019). Trade, migration, and productivity: A quantitative analysis of china. *American Economic Review*, 109(5):1843–1872.
- Xu, Y. and Yang, X. (2021). Access to ports and the welfare gains from domestic transportation infrastructure. *Journal of Urban Economics*, 126:103392.
- Xu, Y. and Yang, X. (2025). Highway construction, labor reallocation, and welfare gains under migration frictions. *Journal of Regional Science*.
- Yuan, W. and Ouyang, D. (2025). Industrial development and trade policy uncertainty: Evidence from china’s wto accession. *Journal of International Economics*, 157:104106.
- Zhang, J. (2011). Interjurisdictional competition for fdi: The case of china’s “development zone fever”. *Regional Science and Urban Economics*, 41(2):145–159.

- Zheng, S. and Kahn, M. E. (2013). China's bullet trains facilitate market integration and mitigate the cost of megacity growth. *Proceedings of the National Academy of Sciences*, 110(14):E1248–E1253.
- Zheng, S., Sun, C., Qi, Y., and Kahn, M. E. (2014). The evolving geography of china's industrial production: Implications for pollution dynamics and urban quality of life. *Journal of Economic Surveys*, 28(4):709–724.
- Zi, Y. (2025). Trade liberalization and the great labor reallocation. *International Economic Review*.

Appendix

A More Evidence on Geography



Figure A1: Map of Chinese Provinces

Figure A2 illustrates the spatial distribution of establishments in two representative sectors. In the early 2000s, foreign investment in electronics primarily flowed into coastal regions. However, by 2005, many electronics firms began relocating inland to capitalize on cheaper labor costs. In contrast, steel companies, which heavily depend on imported raw materials, continued to cluster in coastal areas.

Figure A3 illustrate establishment of particular company groups. For the Foxconn group (one of the largest electronic manufacturing companies in the world), we observed a shift from a coastal dominance to a more balanced distribution between coastal and inland regions. Meanwhile, the distribution of Angang group (China's largest steel manufacturing company) transitioned from a more dispersed pattern to a clear concentration in coastal regions. This evidence underscores the regional and sectoral heterogeneity in the spatial distribution of firms.

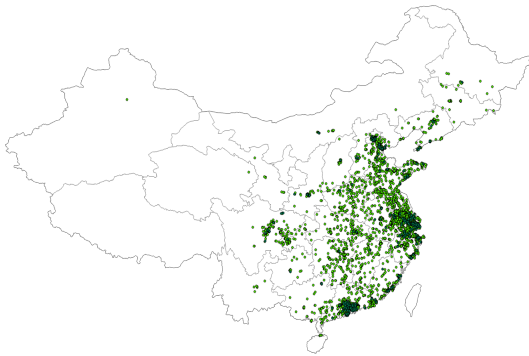
Figure A2: New Entrants



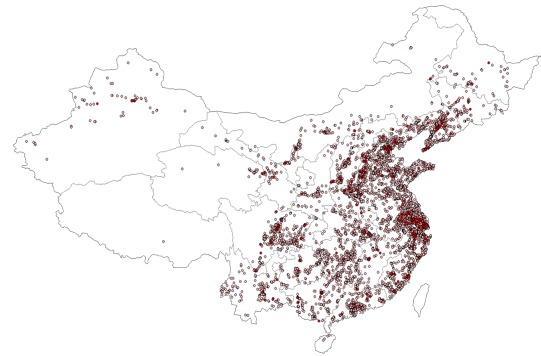
(a) Electronics 1998



(b) Steel 1998



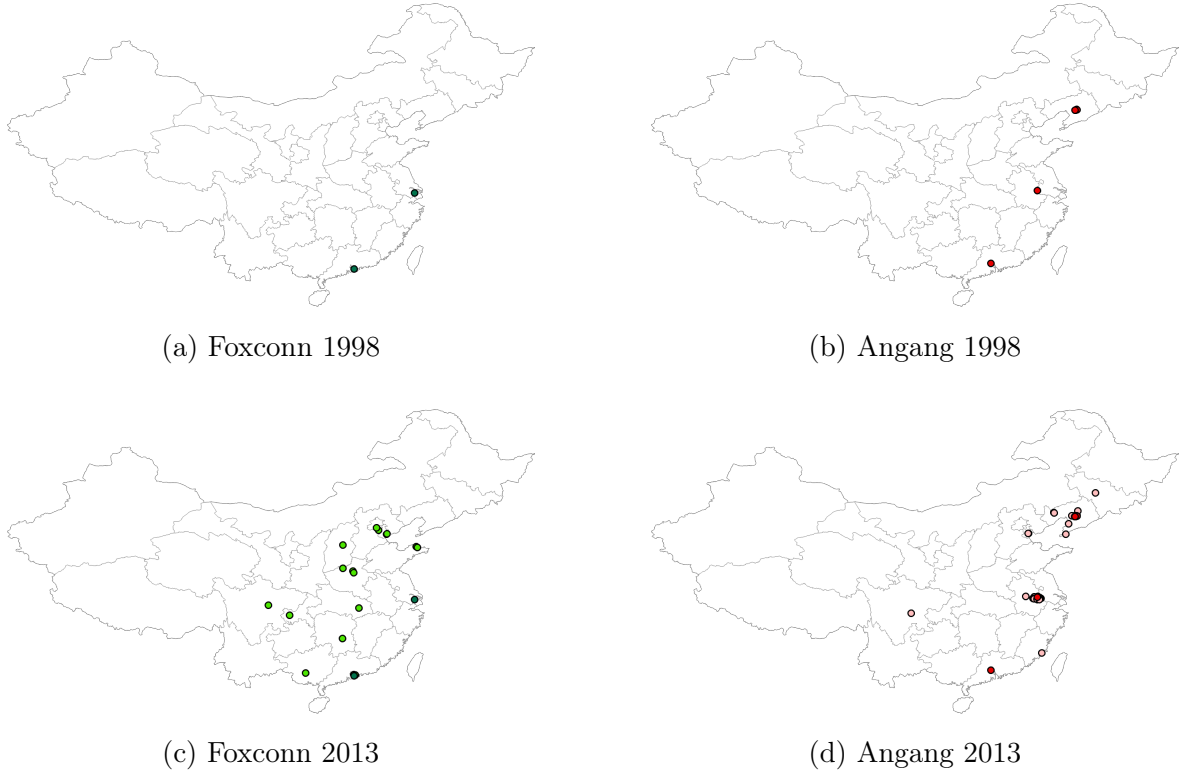
(c) Electronics 2013



(d) Steel 2013

Note: The maps display the spatial distribution of new entrants in 2013 for both the electronics and steel industries. The top left panel shows the electronics industry in 1998, the top right panel shows the steel industry in 1998, the bottom left panel shows the electronics industry in 2013, and the bottom right panel shows the steel industry in 2013. Light colors indicate new entrants in 2013, while dark colors represent establishments that existed in 1998.

Figure A3: Establishment of Particular Company Groups



Note: This figure illustrates the spatial distribution of establishments for Foxconn (electronics) and Angang Group (steel) in both 1998 and 2013. The top left panel shows Foxconn's establishments in 1998, the top right panel shows Angang Group's establishments in 1998, the bottom left panel shows Foxconn in 2013, and the bottom right panel shows Angang Group in 2013. Light colors indicate new establishments in 2013, while dark colors represent establishments from 1998.

B More Evidence on Specialization

We use value add from *MRIO* in 2002 and 2007 to compute the Herfindahl-Hirschman Index (HHI) for each region following $HHI_n = \sum_j (\frac{Y_n^j}{Y_n})^2$, which captures the concentration of the six manufacturing sectors within each region. The HHI is defined as the sum of the squared market shares of all manufacturing sectors, as shown in Table A1. The maximum possible value of HHI is 1, which occurs when a single industry holds 100% of the market share, while all other sectors have a market share of zero. The results in Table A1 suggests that the concentration of manufacturing sectors increased the most in the western region and central region, with an HHI growth of 6.24% and 10.51%, while the eastern region experienced the small reduction of HHI, -1.41%, indicating a relatively stable industrial distribution from 2002 to 2007.

Table A1: HHI

		2002	2007	Δ	$\%\Delta$
Eastern	Mean	0.18	0.19	0.00	2.31%
	Beijing	0.18	0.20	0.02	10.60%
	Tianjin	0.18	0.19	0.02	8.71%
	Hebei	0.20	0.20	0.00	-2.18%
	Shanghai	0.17	0.19	0.01	8.13%
	Jiangsu	0.18	0.18	0.00	-0.85%
	Zhejiang	0.18	0.17	-0.01	-3.19%
	Fujian	0.19	0.19	0.01	3.26%
	Shandong	0.20	0.19	-0.01	-4.17%
	Guangdong	0.18	0.18	0.00	0.45%
Central	Mean	0.20	0.22	0.02	7.01%
	Shanxi	0.28	0.32	0.05	16.51%
	Anhui	0.18	0.18	-0.01	-3.25%
	Jiangxi	0.18	0.21	0.03	18.93%
	Henan	0.20	0.20	0.01	3.66%
	Hubei	0.20	0.20	0.00	-1.87%
	Hunan	0.18	0.19	0.01	8.10%
Northeastern	Mean	0.20	0.21	0.01	4.60%
	Liaoning	0.18	0.19	0.00	1.00%
	Jilin	0.20	0.20	0.00	-0.33%
	Heilongjiang	0.21	0.23	0.03	13.12%
Western	Mean	0.23	0.23	0.01	4.21%
	Inner Mongolia	0.27	0.27	0.00	-1.30%
	Guangxi	0.21	0.20	-0.01	-3.82%
	Chongqing	0.22	0.24	0.01	5.70%
	Sichuan	0.19	0.19	0.00	-0.01%
	Guizhou	0.22	0.26	0.04	18.31%
	Yunnan	0.23	0.24	0.01	3.54%
	Shaanxi	0.18	0.19	0.01	4.35%
	Gansu	0.21	0.23	0.02	9.27%
	Qinghai	0.30	0.28	-0.02	-6.33%
	Ningxia	0.19	0.21	0.02	8.64%
	Xinjiang	0.25	0.27	0.02	7.91%

C More Evidence on Productivity

To observe the differences in regional productivity, we use per capita value-added as a proxy for productivity based on data from the *CIED*. Table A2 presents the disparities in regional productivity between inland and coastal regions. In 2002, productivity was generally higher in coastal regions across most sectors. However, by 2007, the influx of more firms into coastal regions led to a decrease in average productivity, causing coastal regions to show relatively lower productivity in comparison to inland areas in some sectors.

Table A2: Productivity in Inland vs. Coastal Regions

Year	Sector	Inland Mean	Coastal Mean	Inland Median	Coastal Median	Inland Q1	Coastal Q1	Inland Q3	Coastal Q3
2002	Food	71.58	64.77	41.68	36.71	20.75	20.51	86.24	70.78
	Chemicals	79.93	94.10	46.93	54.74	25.30	29.94	89.56	103.13
	Metals	56.31	75.66	33.77	44.14	18.51	24.74	65.74	83.67
	Machinery	62.26	70.84	35.75	42.22	19.98	25.94	67.57	74.22
	Electronics	109.07	79.03	53.97	46.60	26.31	26.08	104.97	87.55
	Others	56.14	53.40	34.25	33.18	18.49	19.83	67.79	61.28
2007	Food	188.45	125.39	105.61	65.86	49.54	36.13	222.03	132.04
	Chemicals	195.22	196.00	110.16	100.63	55.72	52.04	221.47	206.92
	Metals	203.51	169.73	102.72	86.55	49.61	46.29	215.45	178.92
	Machinery	153.60	136.01	85.25	76.07	46.46	44.86	167.79	145.01
	Electronics	212.13	139.20	106.56	67.96	52.66	38.76	216.62	138.30
	Others	149.84	101.47	82.42	56.18	42.64	33.36	160.48	107.09

D Scale Elasticity and Trade Elasticity

Additionally, to introduce scale elasticity and trade elasticity in a consistent manner, we follow the structure outlined in [Kucheryavyi et al. \(2023\)](#). Specifically, by plugging the expression for the productivity cutoff ϕ_{nm}^j into the measure of effective varieties ζ_{nm}^j , we obtain:

$$(\zeta_{nm}^j)^{1-\epsilon^j} = (\tilde{\mu}_n^j)^{\xi^j} (\tilde{b}_m^j)^{\theta^j \xi^j} (T_m^j)^{\xi^j} (c_{nm}^j)^{-\theta^j \xi^j} \left(\frac{w_m^M f_{nm}^j}{X_{nC}^j} \right)^{\frac{\sigma^j - \theta^j - 1}{\sigma^j - 1} \xi^j} (P_{nC}^j)^{\frac{1 - \sigma^j}{(\sigma^j - \theta^j - 1)(1 - \epsilon^j)} \xi^j}.$$

Here, the scale elasticity is defined as $\frac{1}{\theta^j}$, capturing the extent to which average costs fall as the scale of production increases in sector j . A lower θ^j implies stronger increasing returns to scale. The parameter ξ^j plays a central role in governing the model's trade elasticity:

$$\xi^j = \frac{1}{1 + \theta^j \left(\frac{1}{\epsilon^j - 1} - \frac{1}{\sigma^j - 1} \right)}.$$

This expression reflects how both the substitution elasticity across regions (ϵ^j) and across varieties within a region (σ^j), together with the shape of the productivity distribution (θ^j), jointly determine the responsiveness of trade flows to cost shocks. The trade elasticity in this framework is therefore $\xi^j \theta^j$. It reflects the degree of substitutability between products from different regions. High elasticity indicates that consumers can easily switch products, while low elasticity suggests limited substitutability ([Fontagné et al., 2022](#)).

The constant $\tilde{\mu}_n^j$ aggregates key structural parameters and can be written as:

$$\tilde{\mu}_n^j = (\mu^j)^{-\theta^j} \frac{\theta^j}{\theta^j + 1 - \sigma^j} (\sigma^j)^{\frac{\sigma^j - \theta^j - 1}{\sigma^j - 1}}.$$

E Equilibrium Conditions in Hat Algebra

For given structural parameters, θ^j , $\tilde{b}_n^j$, σ^j, ϵ^j , η , α^j , β^j , γ^{ji} , and ρ^i , and for given fundamentals in relative changes, $\{\hat{L}_C\}$, $\{\hat{D}_n\}$, $\{\hat{\tau}_n\}$, $\{\hat{B}_n^j\}$, $\{\hat{f}_{nm}^j\}$, $\{\hat{f}_n^{Ej}\}$, and $\{\hat{A}_n^A\}$, a general equilibrium in relative changes is given by a vector of wages $\{\hat{w}_n^k\}_{n=0,\dots,N}^k$, price indices $\{\hat{P}_{n0}^j\}_{n=0,\dots,N}^j$, $\{\hat{P}_{nC}^j\}_{n=0,\dots,N}^j$, labor allocation $\{\hat{L}_n^k\}_{n=0,\dots,N}^k$, measure of entrants $\{\hat{T}_n^j\}_{n=0,\dots,N}^j$, and cutoff productivity $\{\hat{\phi}_{mn}^j\}_{m,n=0,\dots,N}^j$ that satisfy the following conditions: change in consumer price index implied by (4)

$$\hat{P}_n = \prod_j \left(\hat{P}_n^j \right)^{\alpha^j}, \quad (\text{A.1})$$

change in cost of input bundle implied by (6)

$$\hat{c}_n^j = (\hat{w}_n^M)^{\beta^j} \left(\hat{\Xi}_n^j \right)^{1-\beta^j}, \quad (\text{A.2})$$

change in the cost of composite intermediate inputs implied by (7)

$$\hat{\Xi}_n^j = \prod_{i \in \{1,2,\dots,J\}} \left(\hat{P}_n^i \right)^{\gamma^{ji}}, \quad (\text{A.3})$$

change in marginal cost of serving to region m from region n for firm with $\varphi_m^j = 1$ implied by (8)

$$\hat{c}_{mn}^j = \hat{\tau}_{mn}^j \hat{c}_n^j,$$

change in final good price index implied by (11)

$$\hat{P}_n^j = \left(g_{nC}^j \left(\hat{P}_{nC}^j \right)^{1-\rho^j} + (1 - g_{nC}^j) \left(\hat{P}_{n0}^j \right)^{1-\rho^j} \right)^{\frac{1}{1-\rho^j}}, \quad (\text{A.4})$$

change expenditure share on China-sourced goods implied by (13)

$$\hat{g}_{nC}^j = \frac{\left(\hat{P}_{nC}^j \right)^{1-\rho^j}}{\left(\hat{P}_n^j \right)^{1-\rho^j}}, \quad (\text{A.5})$$

change in expenditure share on RoW-sourced goods implied by (14)

$$\hat{g}_{n0}^j = \frac{\left(\hat{P}_{n0}^j\right)^{1-\rho^j}}{\left(\hat{P}_n^j\right)^{1-\rho^j}}, \quad (\text{A.6})$$

change expenditure share on goods sourced from Chinese region m in total spending on China-sourced goods implied by (15)

$$\hat{\lambda}_{nm}^j = \frac{\left(\hat{\zeta}_{nm}^j\right)^{1-\epsilon^j}}{\left(\hat{P}_{nC}^j\right)^{1-\epsilon^j}}, \quad (\text{A.7})$$

change price index for China-sourced good implied by (17)

$$\hat{P}_{nC}^j = \left(\sum_{m \in \{1,2,\dots,N\}} \lambda_{nm}^j (\hat{\zeta}_{nm}^j)^{1-\epsilon^j} \right)^{\frac{1}{1-\epsilon^j}}, \quad (\text{A.8})$$

change agricultural price index

$$\hat{P}_n^A = \frac{(\hat{w}_n^A)^{\beta^A} (\hat{r}_n)^{1-\beta^A}}{\hat{\mathcal{A}}_n^A},$$

change in cutoff productivity for each pair of regions implied by (19)

$$\hat{\phi}_{nm}^j = \begin{cases} \left(\frac{\hat{w}_m^M \hat{f}_{nm}^j (\hat{c}_{nm}^j)^{\sigma^j-1}}{\hat{X}_{nC}^j (\hat{\zeta}_{nm}^j)^{\sigma^j-\epsilon^j} (\hat{P}_{nC}^j)^{\epsilon^j-1}} \right)^{\frac{1}{\sigma^j-1}} & \text{for } m \neq 0 \\ \left(\frac{\hat{w}_0^M \hat{f}_{n0}^j (\hat{c}_{n0}^j)^{\sigma^j-1}}{\hat{X}_{n0}^j (\hat{P}_{n0}^j)^{\sigma^j-1}} \right)^{\frac{1}{\sigma^j-1}} & \text{for } m = 0 \end{cases}, \quad (\text{A.9})$$

change in average productivity implied by (20)

$$\hat{\phi}_{mn}^j = (\hat{\phi}_{nm}^j) \quad (\text{A.10})$$

change in mass of firms serving from region m to region n implied by (21)

$$\hat{T}_{nm}^j = \left(\frac{\hat{b}_m^j}{\hat{\phi}_{nm}^j} \right)^{\theta^j} \hat{T}_m^j \quad (\text{A.11})$$

change in price index of goods sourced from Chinese region m and sold in n implied by (22)

$$\hat{\zeta}_{nm}^j = \frac{\left(\hat{T}_{nm}^j\right)^{\frac{1}{1-\sigma^j}} \hat{c}_{nm}^j}{\hat{\phi}_{nm}^j} \quad \text{for } m \neq 0. \quad (\text{A.12})$$

change in price index of goods sourced from the RoW implied by (23)

$$\hat{P}_{n0}^j = \frac{\left(\hat{T}_{n0}^j\right)^{\frac{1}{1-\sigma^j}} \hat{c}_{n0}^j}{\hat{\phi}_{n0}^j} \quad (\text{A.13})$$

change in total spending implied by (24)

$$\hat{X}_n^j = \frac{HE_n^j}{X_n^j} HE_n^j + \frac{ME_n^j}{X_n^j} \hat{ME}_n^j, \quad (\text{A.14})$$

change in total spending on China-sourced goods implied by (25)

$$\hat{X}_{nC}^j = \hat{g}_{nC}^j \hat{X}_n^j,$$

total spending on RoW-sourced goods (26)

$$\hat{X}_{n0}^j = \hat{g}_{n0}^j \hat{X}_n^j,$$

change in total sales from m to n implied by (27)

$$\hat{X}_{nm}^j = \begin{cases} \hat{\lambda}_{nm}^j \hat{X}_{nC}^j & \text{for } m \neq 0 \\ \hat{X}_{n0}^j & \text{for } m = 0 \end{cases}, \quad (\text{A.15})$$

change in total sales of firms in region n implied by (28)

$$\hat{Y}_n^j = \sum_{m=0,1,\dots,N} \frac{X_{mn}^j}{Y_n^j} \hat{X}_{mn}^j, \quad (\text{A.16})$$

change in total spending on intermediate input from sector j implied by (29)

$$\hat{ME}_n^j = \sum_{i=1}^J \frac{ME_n^{ij}}{ME_n^j} \hat{Y}_n^i, \quad (\text{A.17})$$

change in mass of entrants implied by (31)

$$\hat{T}_m^j = \frac{\hat{Y}_m^j}{\hat{w}_m^M \hat{f}_m^j E},$$

change in workers spending implied by (34)

$$\hat{H}E_n = \frac{L_n^M w_n^M}{L_n^M w_n^M + L_n^A w_n^A} \hat{L}_n^M \hat{w}_n^M + \frac{L_n^A w_n^A}{L_n^M w_n^M + L_n^A w_n^A} \hat{L}_n^A \hat{w}_n^A, \quad (\text{A.18})$$

change in worker sectoral spending implied by (35)

$$\hat{H}E_n^j = \hat{H}E_n, \quad (\text{A.19})$$

change in worker's location choice probability implied by (32)

$$\hat{\psi}_n^M = \frac{\hat{B}_n^M \left(\frac{\hat{w}_n^M}{\hat{P}_n} \right)^\eta}{\sum_m \psi_m^k \hat{B}_m^k \left(\frac{\hat{w}_m^k}{\hat{P}_m} \right)^\eta}, \quad (\text{A.20})$$

$$\hat{\psi}_n^A = \frac{\hat{B}_n^A \left(\frac{\hat{w}_n^A}{\hat{P}_n} \right)^\eta}{\sum_m \psi_m^k \hat{B}_m^k \left(\frac{\hat{w}_m^k}{\hat{P}_m} \right)^\eta}, \quad (\text{A.21})$$

change in mass of manufacturing and agricultural workers in each region in China implied by (33)

$$\hat{L}_n^M = \hat{\psi}_n^M \hat{L}_C \quad \text{for } n \neq 0, \quad (\text{A.22})$$

$$\hat{L}_n^A = \hat{\psi}_n^A \hat{L}_C \quad \text{for } n \neq 0, \quad (\text{A.23})$$

change in land rent implied by (38)

$$\hat{\hat{r}}_n = \frac{\hat{\hat{E}}_n^A}{\hat{\hat{D}}_n}, \quad (\text{A.24})$$

change in wages implied by (36)

$$\hat{w}_n^M = \frac{\sum_j \frac{\left(\frac{\beta^j (\sigma^j - 1) + 1}{\sigma^j} \right) Y_n^j}{\sum_j \left(\frac{\beta^j (\sigma^j - 1) + 1}{\sigma^j} \right) Y_n^j} \hat{Y}_n^j}{\hat{L}_n^M}, \quad (\text{A.25})$$

change in agricultural wages implied by (37)

$$\hat{w}_n^A = \frac{\hat{E}_n^A}{\hat{L}_n^A}, \quad (\text{A.26})$$

and change in welfare implied by (39)

$$\hat{W}_C = \left(\sum_n \sum_k \psi_n^k \hat{B}_n^k \left(\frac{\hat{w}_n^k}{\hat{P}_n} \right)^\eta \right)^{\frac{1}{\eta}}.$$

F Solution Algorithm

We use the following order to solve the model. The baseline equilibrium variables g_{nC}^j , λ_{nm}^j , $\frac{HE_n^j}{X_n^j}$, $\frac{ME_n^j}{X_n^j}$, $\frac{ME_n^{ij}}{ME_n^j}$, $\frac{X_{nm}^j}{Y_n^j}$, $\frac{Y_n^j}{\sum_j Y_n^j}$, and ψ_n^k , and parameters θ^j , $\tilde{b}_n^j$, σ^j, ϵ^j , η , α^j , β^j , γ^{ji} , and ρ^i are given.

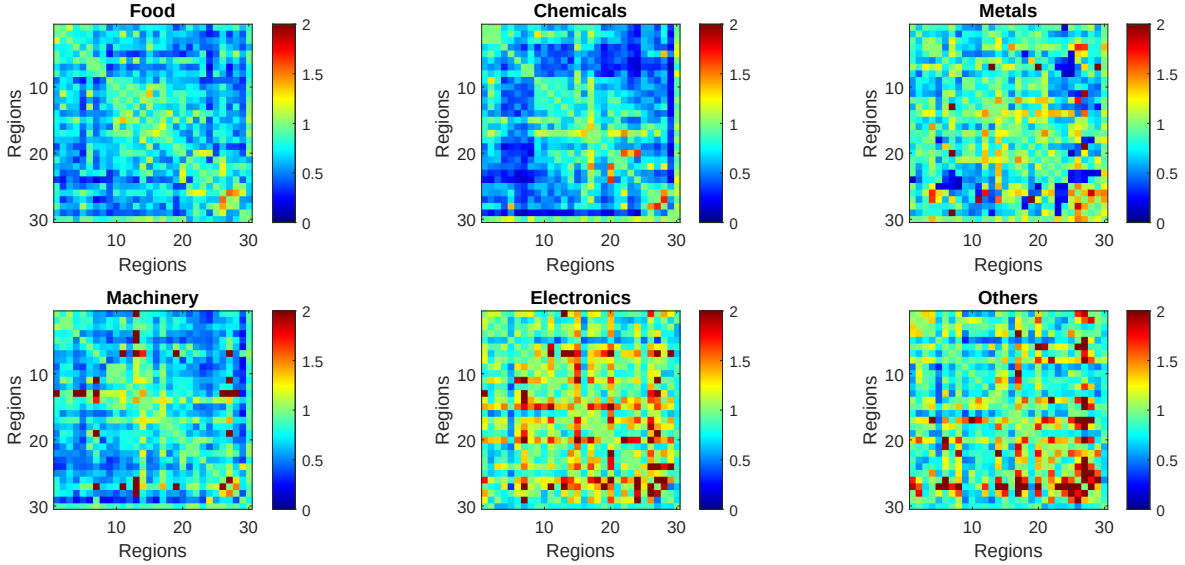
1. Guess productivity cutoff $\{\hat{\phi}_{mn}^j\}_{mn}^j$, wage $\{\hat{w}_n^k\}_n^k$, mass of firms $\{\hat{T}_n^j\}_n^j$, and output $\{\hat{Y}_n^j\}_n^j$ implied by (A.10), (A.11), and determine household expenditure $\{\hat{H}E_n\}_n$, $\{\hat{H}E_n^j\}_n^j$, land rent $\{\hat{r}_n\}_n$ implied by (A.18), (A.19), (A.24).
2. Solve price index $\{\tilde{P}_n^j\}_n^j$
 - 2.1 Determine final good price index $\{\hat{P}_n^j\}_n^j$, expenditure share on RoW-sourced goods $\{\hat{g}_{n0}^j\}_n^j$, expenditure share on China-sourced goods $\{\hat{g}_{nC}^j\}_n^j$ implied by (A.4), (A.5), (A.6).
 - 2.2 Determine final good price index composite price index of intermediate input $\{\hat{\Xi}_n^j\}_n^j$, unit cost $\{\hat{c}_n^j\}_n^j$, implied by (A.3), (A.2).
 - 2.3 Update price index of goods sourced from Chinese region $\{\hat{\zeta}_{nm}^j\}_{nm}^j$, price index $\{\tilde{P}_{nC}^j\}_{nC}^j$, $\{\tilde{P}_{n0}^j\}_{n0}^j$, and expenditure share on goods sourced from Chinese region $\{\hat{\lambda}_{nm}^j\}_{nm}^j$ implied by (A.12), (A.8), (A.13), (A.7).
3. Determine household price index $\{\hat{P}_n\}_n$, rural-urban location choice $\{\hat{\psi}_n^k\}_n^k$, number of workers in each region $\{\hat{L}_n^k\}_n^k$, household expenditure $\{\hat{H}E_n\}_n$, $\{\hat{H}E_n^j\}_n^j$, demand for each sectoral input $\{\hat{M}E_n^j\}_n^j$, sales $\{\hat{X}_{nm}^j\}_{nm}^j$, and spending $\{\hat{X}_n^j\}_n^j$ implied by (A.1), (A.20), (A.21), (A.22), (A.23), (A.18), (A.19), (A.17), (A.15), and (A.14).
4. Update total sales $\{\hat{Y}_n^j\}_n^j$ implied by (A.16).
5. Update cutoff productivity $\{\hat{\phi}_{mn}^j\}_{mn}^j$ implied by (A.9).
6. Update wage $\{\hat{w}_n^k\}_n^k$ implied by (A.25) and (A.26).

G Descriptive Statistics of Shocks

Table A3: Chang in Sunk Entry Costs between 2002 and 2007

	Food	Chemicals	Metals	Machinery	Electronics	Others
Beijing	-59.74%	-50.47%	-50.13%	-31.47%	-57.48%	-72.95%
Tianjin	-40.85%	-33.76%	-52.27%	-44.85%	-52.46%	-38.68%
Hebei	-49.35%	-41.99%	-27.54%	-56.72%	0.31%	-38.20%
Shanxi	-88.15%	-57.80%	-56.12%	-82.93%	-76.51%	-91.85%
Inner Mongolia	-81.28%	-78.28%	-77.78%	-83.77%	-83.00%	-91.45%
Liaoning	-66.34%	-45.89%	-57.57%	-51.11%	-45.54%	-49.59%
Jilin	-74.10%	-61.61%	-74.18%	-28.09%	-78.96%	-73.99%
Heilongjiang	-52.29%	134.71%	287.44%	-12.86%	13.43%	-26.53%
Shanghai	-26.01%	-9.52%	-26.90%	-14.52%	34.56%	-41.68%
Jiangsu	-61.50%	-52.95%	-49.95%	-50.99%	-55.13%	-57.46%
Zhejiang	-51.57%	-29.11%	-45.45%	-13.09%	74.18%	-31.28%
Anhui	-59.67%	-32.93%	-64.00%	-58.32%	48.52%	-69.53%
Fujian	-21.65%	-4.50%	-11.00%	-50.77%	0.40%	-56.78%
Jiangxi	-75.69%	-62.72%	-55.93%	-69.94%	-41.57%	-68.43%
Shandong	-46.20%	-54.13%	-58.12%	-50.65%	-8.52%	-56.21%
Henan	-60.17%	-35.50%	-48.06%	-52.28%	-28.39%	-62.07%
Hubei	-63.90%	-56.61%	-70.44%	-64.57%	-44.74%	-61.62%
Hunan	-67.81%	-47.46%	-60.77%	-58.11%	21.76%	-61.74%
Guangdong	-38.21%	-44.32%	-22.25%	-30.43%	-65.40%	-58.34%
Guangxi	73.49%	-42.96%	-2.22%	-19.34%	115.75%	23.13%
Chongqing	-72.21%	-69.33%	-73.61%	-64.88%	-84.50%	-78.99%
Sichuan	-58.70%	-58.97%	-70.10%	-34.53%	-47.81%	-77.36%
Guizhou	-50.33%	-42.80%	-71.67%	-79.41%	-47.10%	-61.93%
Yunnan	-11.48%	-47.08%	-8.45%	-13.79%	26.55%	-51.78%
Shaanxi	-80.78%	-26.84%	-35.54%	-50.30%	-54.56%	-85.00%
Gansu	-87.31%	-73.15%	-78.94%	-92.12%	-74.97%	-90.75%
Qinghai	-88.36%	-36.98%	-59.60%	-76.49%	-97.24%	-67.51%
Ningxia	-71.22%	-63.46%	-85.33%	-82.27%	-56.66%	-75.96%
Xinjiang	-82.04%	-30.71%	-60.78%	-94.00%	-35.62%	-78.38%
Mean	-55.63%	-39.90%	-40.25%	-52.16%	-27.61%	-60.45%

Figure A4: Chang in Trade Cost Ratio between 2002 and 2007



Note: The values in the heatmap indicate the relative changes in iceberg trade costs from 2007 to 2002, $\tau_{nm,2007}^j / \tau_{nm,2002}^j$.

H Model Fit

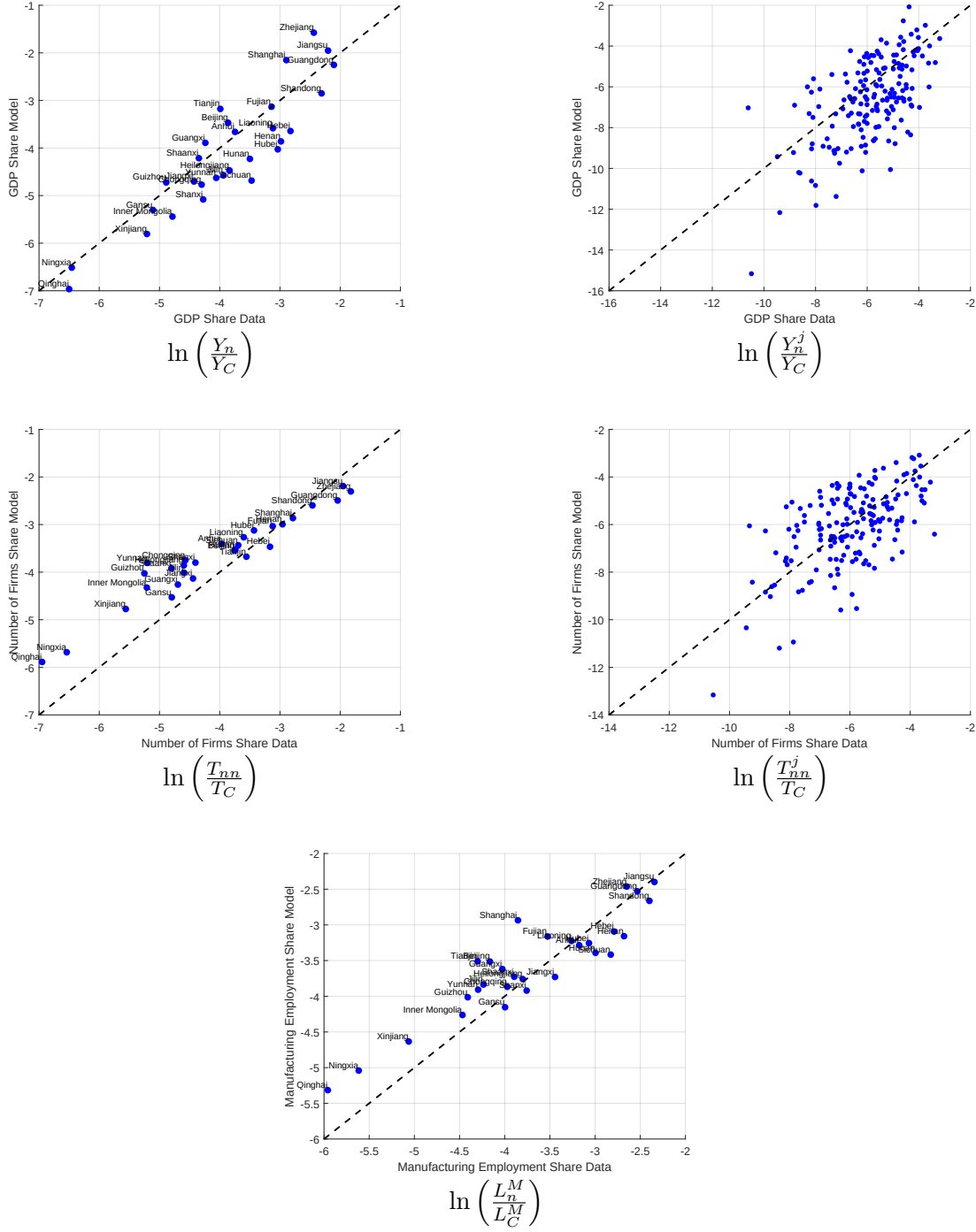
Here, we present the simulation results of our model. We first solve model in level and then compare the actual economic situation of China in 2002 with the results simulated by our model. Table A4 summarizes the method of calibration in both level and hat.

After solving model in level, we then solve model in hat. By introducing shocks (employment, trade costs, and sunk entry costs) to the baseline year 2002, we assess how closely the simulated outcomes align with the actual data from 2007. While we cannot capture every shock, we believe the three shocks (to trade costs, sunk entry costs, and labor) played a significant role in driving China's economic development. The results are shown in Figure A5. The x-axis represents the observed data, while the y-axis shows the simulated results of the model.

Table A4: Summary of Calibration

Parameters	Solve Model in	
	Level	Hat
α^j		WIOD
β^j		WIOD
β^A		Tombe and Zhu (2019)
γ^{ij}		WIOD
σ^j		CIED
θ^j		CIED
ϵ^j		Bartelme et al. (2025)
ρ^j		Caliendo et al. (2023)
η^j		CIED
B_n^k		MRIO, Survey Data
$\tilde{b}_n^j$		1
$\mathcal{A}_n^A$		1
D_n	$\frac{1}{N-1}$	1
Baseline Outcomes		
g_{nC}^j	/	MRIO
λ_{nm}^j	/	MRIO
$\frac{HE_n^j}{X_n^j}$	/	MRIO
$\frac{ME_n^j}{X_n^j}$	/	MRIO
$\frac{ME_n^{kj}}{ME_n^j}$	/	MRIO
$\frac{X_{nm}^j}{Y_n^j}$	/	MRIO
ψ_n^k	/	Survey Data
Shocks to Fundamentals		
L_C		Survey Data
L_0^A		OECD
L_0^M		OECD
τ_{nm}^j		MRIO, WIOD, Custom Data
f_n^{Ej}		MRIO, CERD
f_{nm}^j		1

Figure A5: “Fit” of the Model: Level



I Decomposition of Real Wage by Covariate Contributions

Table A5: Decomposition of Real Wage R^2 by Covariate Contributions

	Term 1	Term 2	Term 3	Term 4	Term 5	Share of Contributions		
						“Generic” Terms w/o 3	“Selection” Term 3	R^2
Shock: All								
Domin. Stat. Ranking	0.371 2	0.038 3	0.488 1	0.030 4	0.024 5	0.464	0.488	0.952
Shock: L								
Domin. Stat. Ranking	0.251 2	0.118 4	0.376 1	0.104 5	0.141 3	0.614	0.376	0.991
Shock: $L + \tau^{Dom}$								
Domin. Stat. Ranking	0.300 2	0.060 5	0.359 1	0.066 4	0.107 3	0.532	0.359	0.890
Shock: $L + \tau^{Int}$								
Domin. Stat. Ranking	0.401 1	0.106 3	0.259 2	0.106 4	0.065 5	0.678	0.259	0.937
Shock: $L + \tau^{Both}$								
Domin. Stat. Ranking	0.397 1	0.058 5	0.203 2	0.065 4	0.136 3	0.656	0.203	0.859
Shock: $L + f^E$								
Domin. Stat. Ranking	0.306 2	0.100 3	0.458 1	0.092 4	0.025 5	0.523	0.458	0.981

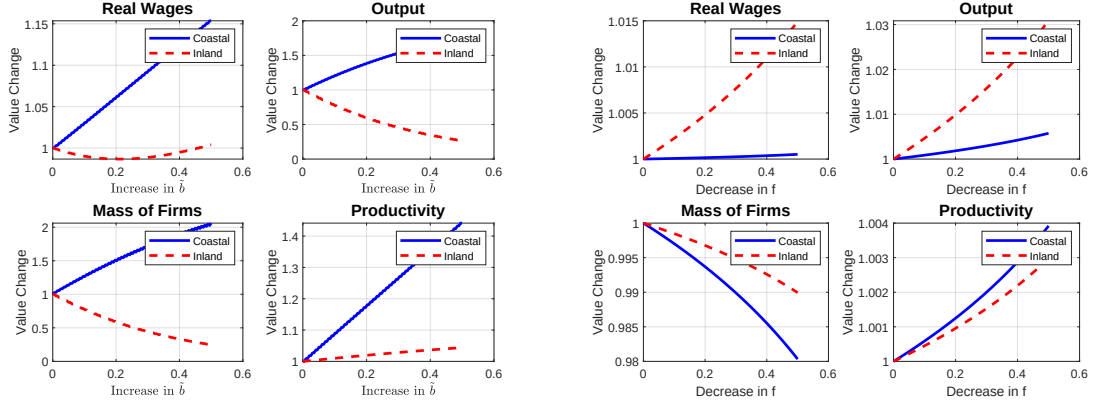
Note: This dominance analysis decomposes the total R^2 to quantify each explanatory variable’s relative contribution to explaining variations in real wages.

J Exploring How Fixed Costs, Sunk Entry Costs, and Pareto Location Parameter Affect Average Productivity: A Numerical Simulation Study

In order to do the simple analysis, we modify our model into a toy model by dividing it into two symmetric sectors—coastal and inland regions. We then reduce the export fixed costs to other areas, decrease the sunk entry costs, and increase the Pareto location parameter, but only for the two coastal-sector sectors, while keeping all other values unchanged. We observe that each of these changes affects the selection effect, meaning that the average productivity in the coastal region increases.

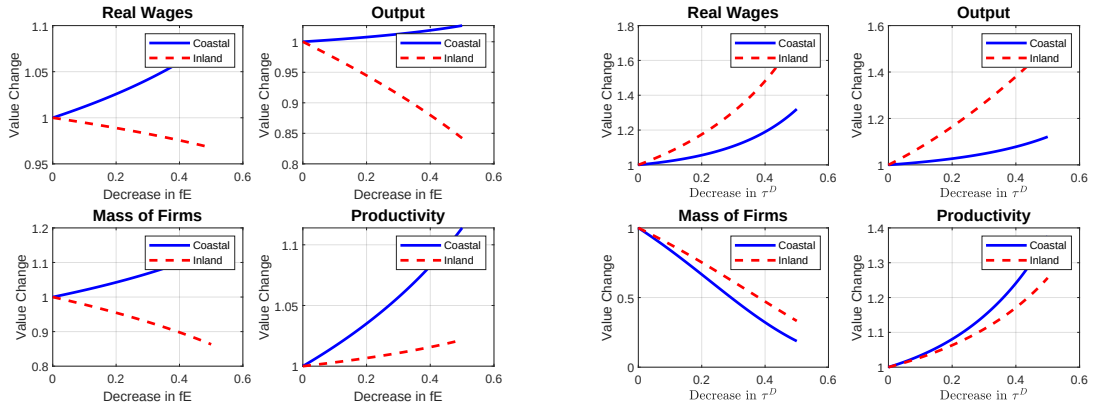
In the main text, due to data limitations, we are unable to calibrate the fixed costs and the Pareto location parameter. Therefore, we assume that in our model the selection effect arises from shocks other than the fixed costs and the Pareto location parameter.

Figure A6: Numerical Exercise



Change in Pareto Location Parameter

Change in Fixed Costs



Change in Sunk Entry Costs

Change in Domestic Trade Costs

Note: These figures show how real wages $\frac{w_n^M}{P_n^M}$, output Y_n^j , mass of firms T_{nn}^j , and productivity cutoff ϕ_{nn}^j affected by change in Pareto location parameter $\tilde{b}_1^j$, fixed costs f_{12}^j, f_{21}^j , sunk entry costs f_1^{Ej} , and domestic trade costs τ_{12}^j, τ_{21}^j .

K SOE

Due to the differences in stages between state-owned enterprises (SOEs) and private enterprises, our model can distinguish these differences. We assume that SOE status is exogenously determined by the government, and SOEs face lower entry costs, which are partly paid by the government and shared by all households based on the soft budget constraint (Kornai et al., 2003; Qian and Roland, 1998). This implies that SOEs have a lower entry cutoff. As a result, even if SOEs draw productivity from the same Pareto distribution as private enterprises, they tend to have lower average productivity. This can be supported by empirical evidence as well as casual observation. Most importantly, this assumption allows us to conduct a counterfactual analysis by examining the gains from “hardening” the budget constraints for SOEs, making them face the same entry costs as private enterprises.

Based on the above setting, this section presents the empirical analysis conducted to examine the relationships outlined in our theoretical model. Utilizing comprehensive datasets and robust econometric techniques, we investigate how distance to port influences firms’ import and export within China’s steel and electronics sector.

Data

Our empirical analysis draws on multiple data sources to comprehensively examine the dynamics of China’s steel industry. The primary dataset is the *CIED*, which includes detailed firm-level data on steel and electronics establishments. In addition to the *CIED*, we incorporate data from *China’s Customs Database*, particularly focusing on detailed import data of iron ore or semiconductor, which is crucial for steel production.

To capture the dynamic nature of trade costs over time, we construct a time-varying weighted distance metric, which accounts for the variation in the import sources and trade costs faced by firms in different regions.

Constructing Time-Varying Weighted Distance

To quantify the trade costs faced by firms in sourcing imports, we develop a time-varying weighted distance metric. This metric calculates the effective distance between a firm’s location and its import sources, weighted by the share of imports coming from each source. Formally, this is expressed as:

$$\hat{\text{Distance}}_{int} = \sum_{m=1}^M [\text{Distance}_{nm} \times \text{Import Share}_{nmt}]$$

where Distance_{nm} represents the distance between firm n and port m . $\text{Import Share}_{nmt}$ denotes the share of imports from port m by firm n at time t . This metric effectively captures the

weighted average distance based on the actual import sources of each firm, thereby providing a nuanced measure of trade costs that evolves over time. We estimate several regression models to explore the impact of distance on different dependent variables: imported input share, export decisions, export value, and wages. Each model incorporates relevant control variables and fixed effects to isolate the effect of distance.

Imported Input Share

The theoretical model suggests that the expenditure share of sector j 's input from abroad is determined by the relative prices of domestic and foreign inputs. This relationship is formalized as:

$$g_{n0}^j = \frac{\left(\tilde{P}_{n0}^j\right)^{1-\rho^j}}{\left(\tilde{P}_{nC}^j\right)^{1-\rho^j} + \left(\tilde{P}_{n0}^j\right)^{1-\rho^j}}$$

where $\tilde{P}_{n0}^j = \tau_{n0}^j \frac{\mu^j (T_{n0}^j)^{\frac{1}{1-\sigma^j}} c_0^j}{\phi_{n0}^j}$. Based on this, we specify the following regression model to estimate the imported input share:

$$\frac{\text{Import}_{int}}{\text{Import}_{int} + \text{Domestic Input}_{int}} = \mathbb{A} - \mathbb{B} \ln(\widehat{\text{Distance}}_{int}) + \delta X_{int} + \psi_i + \nu_t + \varepsilon_{int}$$

where $\ln(\widehat{\text{Distance}}_{int})$ is the logged time-varying weighted distance, and X_{int} includes control variables such as firm capital, age, debt ratio, and a dummy for SOEs. Year-fixed and firm-fixed effects control for time-specific shocks and unobserved, time-invariant differences between firms.

Table A6 provides a detailed view of the relationship between distance and the imported input share for the steel and electronics sectors. The results show a significant impact of distance on the imported input share for non-SOE firms in both sectors. Specifically, in the steel sector, greater distances are associated with a significant decrease in the imported input share, as indicated by the negative and statistically significant coefficients for the overall sample and non-SOE firms. This suggests that increased trade costs, reflected by longer distances, constrain non-SOE firms' ability to source inputs from abroad, which is consistent with the notion that non-SOE firms are more sensitive to trade costs.

However, for SOE firms in the steel sector, the coefficient for distance is not statistically significant, highlighting the influence of soft budget constraints that may shield these firms from the trade cost pressures faced by non-SOE firms. The lack of significance for SOEs indicates that institutional factors, such as access to resources and government support, likely

mitigate the trade costs associated with distance.

In the electronics sector, the results for non-SOE firms similarly show a negative but insignificant coefficient for distance, while the coefficient for SOE firms is positive, though also statistically insignificant. This difference between sectors may reflect varying sensitivities to distance due to sector-specific factors, such as the types of inputs required or differences in infrastructure and logistics support.

Table A6: Imported Input Share Regression Results

	All: 2000-2007	SOE	Non-SOE
Steel	Imported Input Share		
ln(Distance)	-0.0004*** (-3.22)	-0.0005 (-0.19)	-0.0004*** (-3.13)
Controls	✓	✓	✓
Year FE	✓	✓	✓
Firm FE	✓	✓	✓
N	40,644	2,217	38,427
Electronics	Imported Input Share		
ln(Distance)	-0.0001 (-1.00)	0.001 (1.39)	-0.0001 (-1.13)
Controls	✓	✓	✓
Year FE	✓	✓	✓
Firm FE	✓	✓	✓
N	48,108	2,497	45,61
Notes: t statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$			

Export Decision

A firm's decision to export is modeled based on the condition that the cost of exporting is outweighed by the expected revenue from foreign markets. Mathematically, this is expressed as:

$$f_{0n}^j w_n < \frac{1}{\sigma^j} r_{n0}^j (\varphi_0^j).$$

We set the regression model which is similar to the imported input share regression, controls for firm characteristics, year-fixed effects, and SOE status are included:

$$\text{Export Dummy}_{int} = \mathbb{A} - \mathbb{B} \ln(\text{Distance}_{int}) + \delta X_{int} + \nu_t + \varepsilon_{int}$$

Table A7 illustrates the relationship between distance and the likelihood of firms engaging in export activities for the steel and electronics sectors. In the steel sector, the regression results show a negative and statistically significant coefficient for distance in the overall sample and for non-SOE firms. This finding suggests that greater distances, and thus higher trade costs,

significantly reduce the probability of non-SOE firms exporting their products, consistent with the idea that increased distance imposes higher barriers to entry into international markets for these firms.

For SOE firms in the steel sector, however, the coefficient for distance is positive but not statistically significant, indicating that distance does not have a significant deterrent effect on their export decisions. This could be due to the institutional advantages enjoyed by SOE firms, such as government support, which may mitigate the negative impact of trade costs associated with longer distances.

In the electronics sector, the results follow a similar pattern, with non-SOE firms showing a negative but insignificant coefficient for distance, while SOE firms display a positive yet insignificant coefficient. The lack of statistical significance in the electronics sector may be attributed to sector-specific factors, such as greater logistical efficiencies or the nature of the products being exported, which may reduce the sensitivity to distance compared to the steel sector.

Table A7: Export Decision Regression Results: Steel

	All: 2000-2007	SOE	Non-SOE
Steel	Export Dummy		
ln(Distance)	-0.035*** (-2.97)	0.035 (0.97)	-0.040*** (-3.14)
Controls	Yes	Yes	Yes
Year Dummy	Yes	Yes	Yes
Observations	41,071	2,438	38,633
Electronics	Export Dummy		
ln(Distance)	-0.0001 (-1.00)	0.001 (1.39)	-0.0001 (-1.13)
Controls	Yes	Yes	Yes
Year Dummy	Yes	Yes	Yes
N	48,108	2,497	45,61
Notes: t statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$			

The empirical results provide strong evidence that trade costs significantly shape firms' input sourcing and export behaviors in China's steel and electronics industries. Non-SOE firms are particularly sensitive to trade costs, as longer distances reduce their reliance on imported inputs and lower their likelihood of exporting. This highlights the need for reducing trade costs to enhance the competitiveness of non-SOE firms in global markets. In contrast, SOE firms appear less affected by distance, likely due to institutional support that helps cushion the impact of higher trade costs. These findings suggest that trade liberalization and infrastructure improvements have varying effects depending on firms' ownership structures.

L Full Counterfactual Results

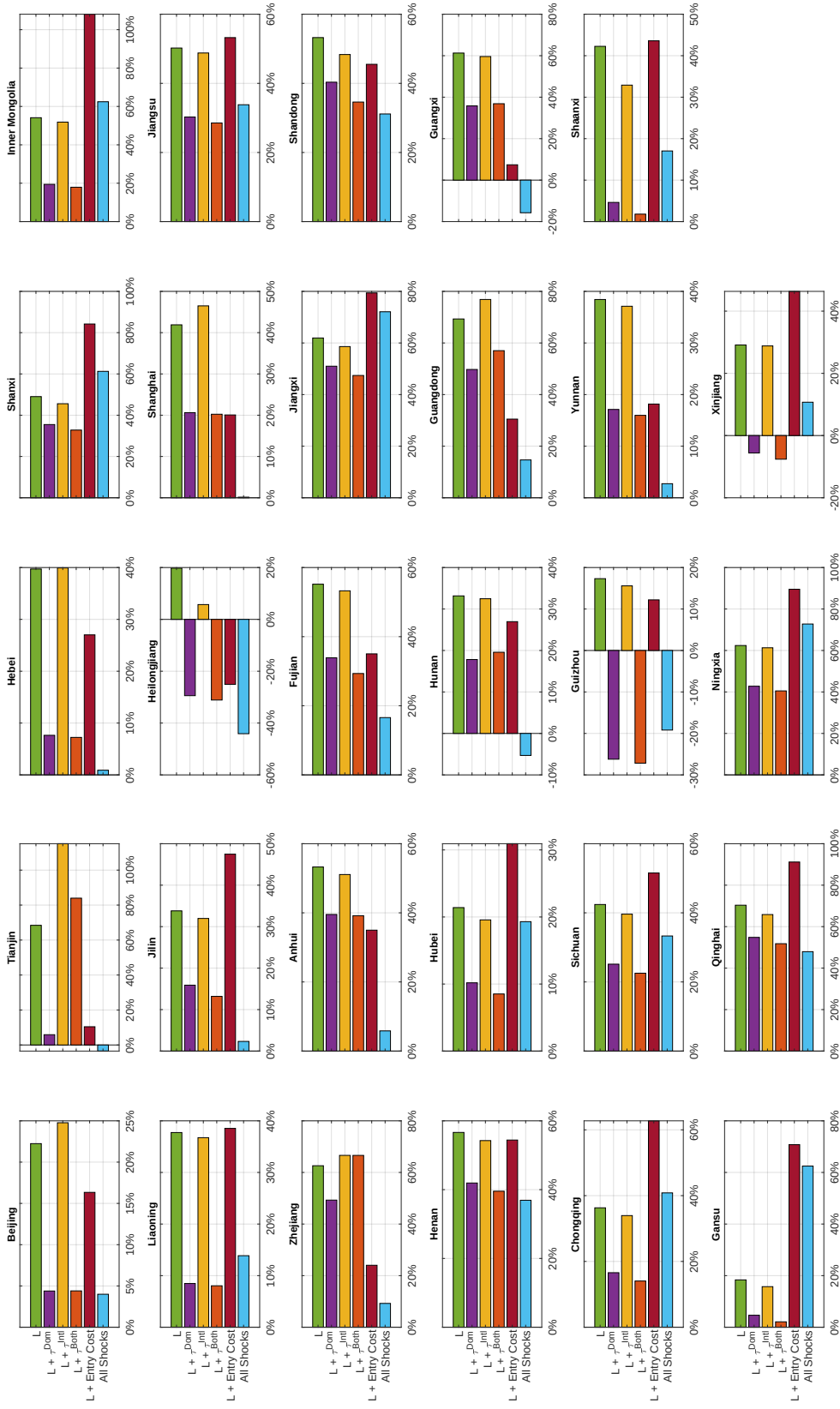


Figure A7: Change of Number of Firms

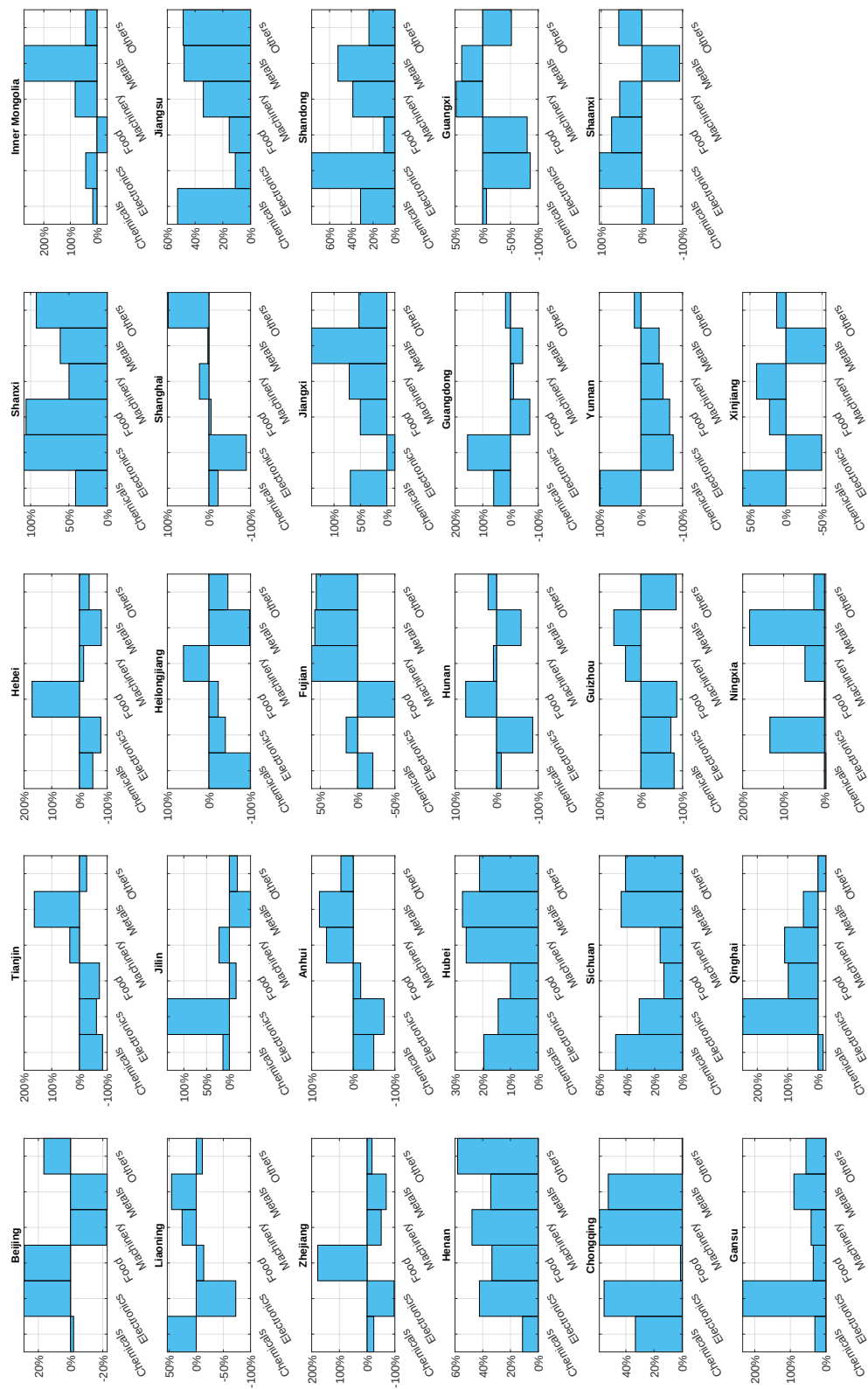


Figure A8: Change of Sectoral Number of Firms

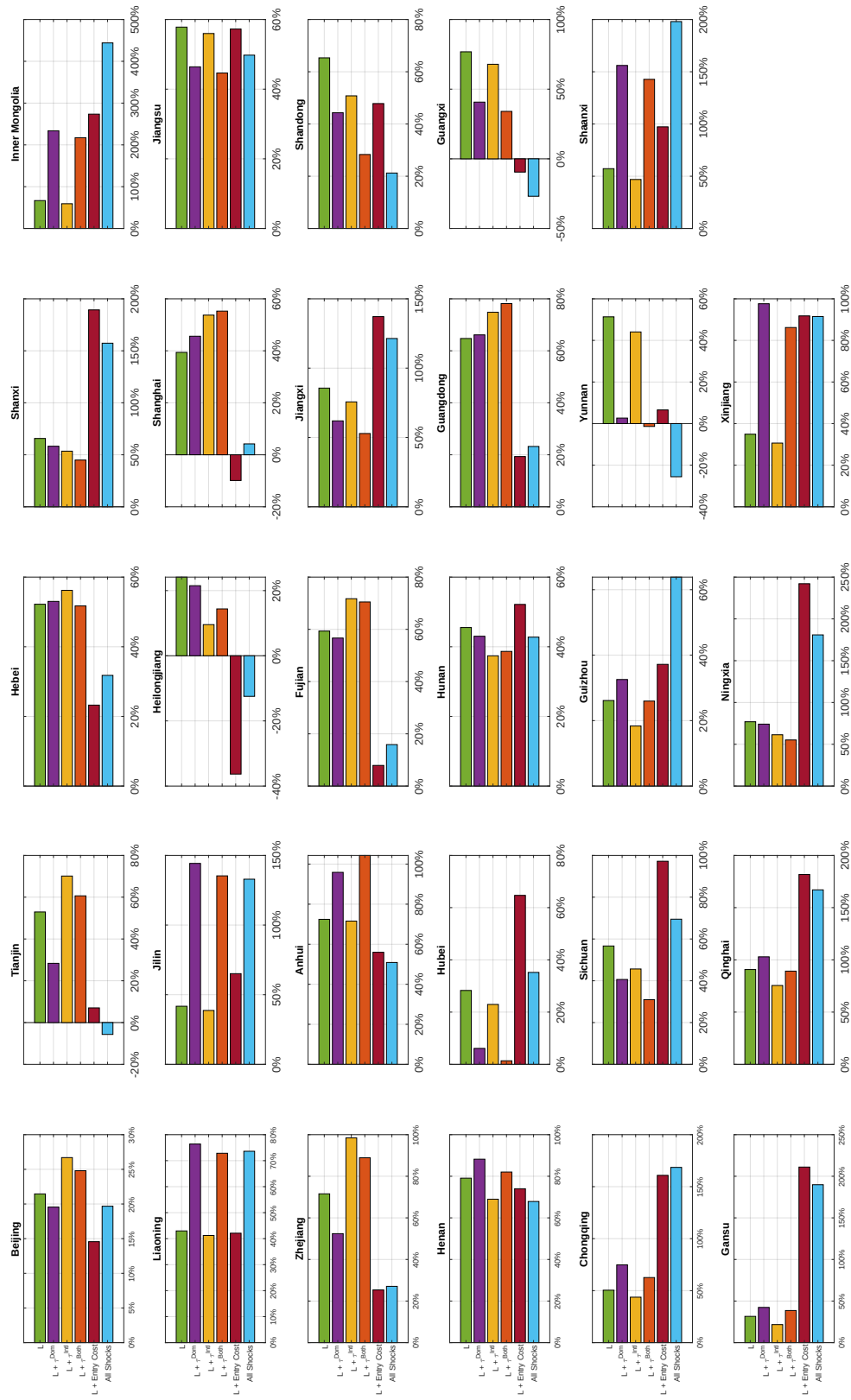


Figure A9: Change of GDP

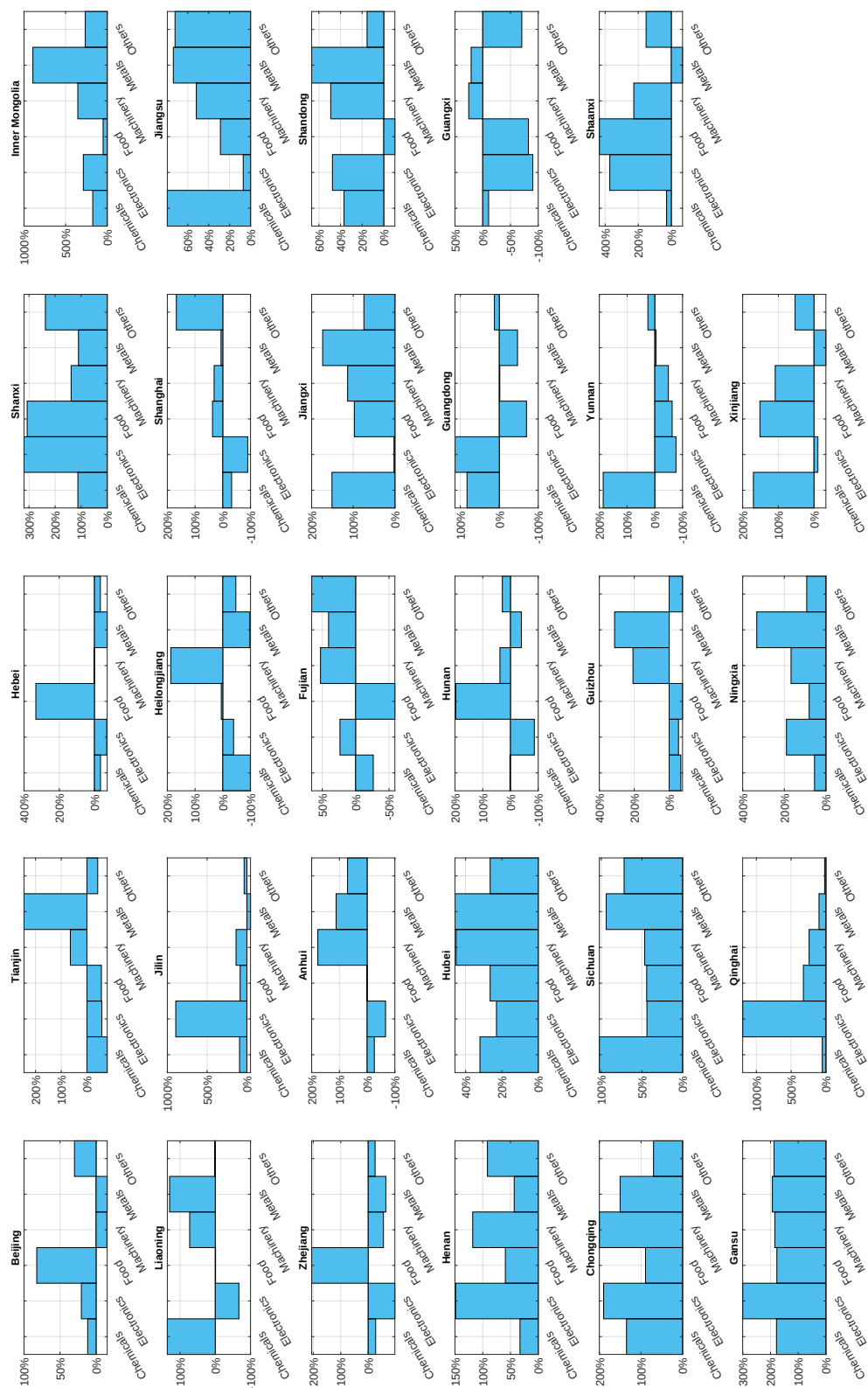


Figure A10: Change of Sectoral GDP

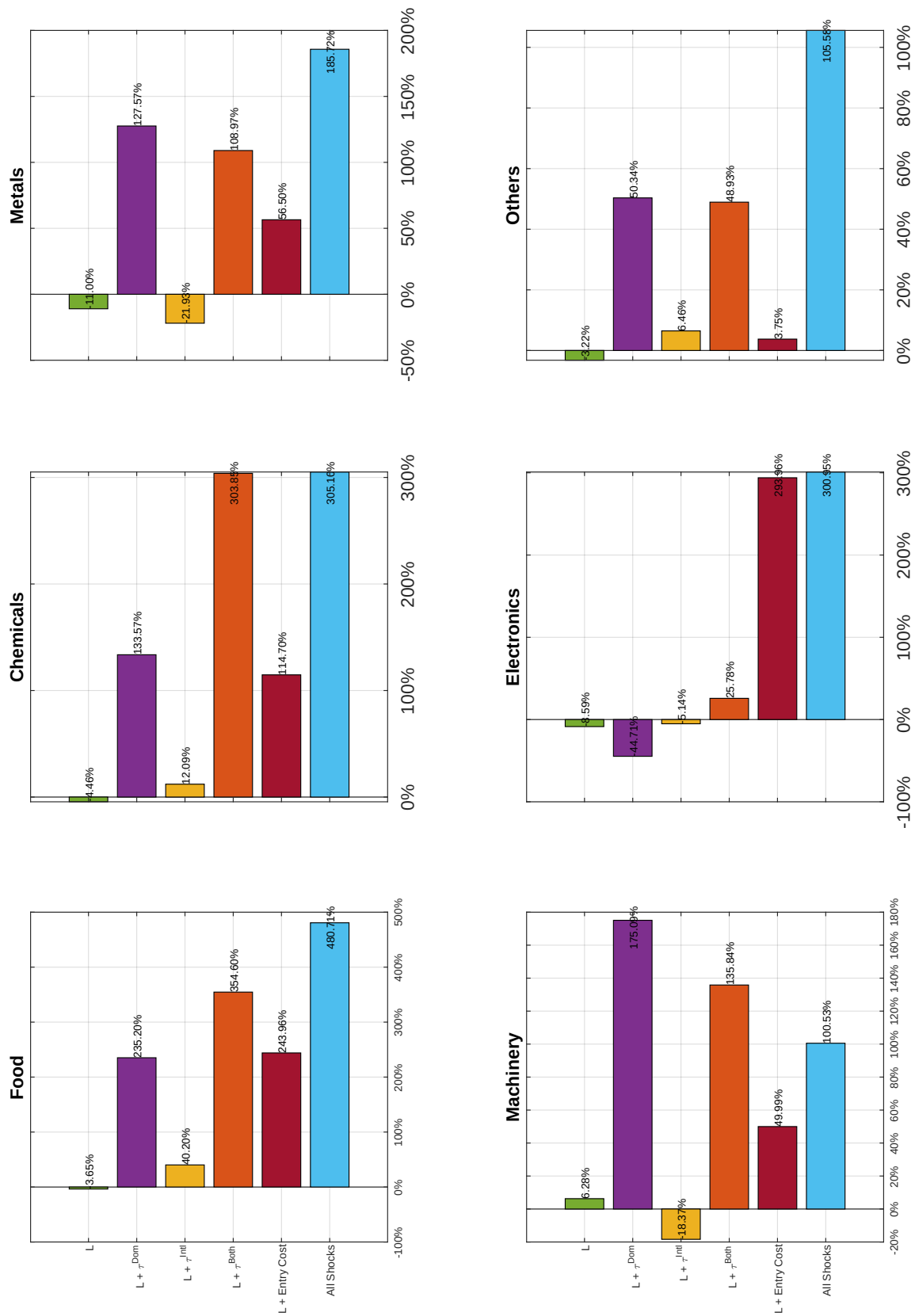


Figure A11: Change of Sectoral Specialization Index

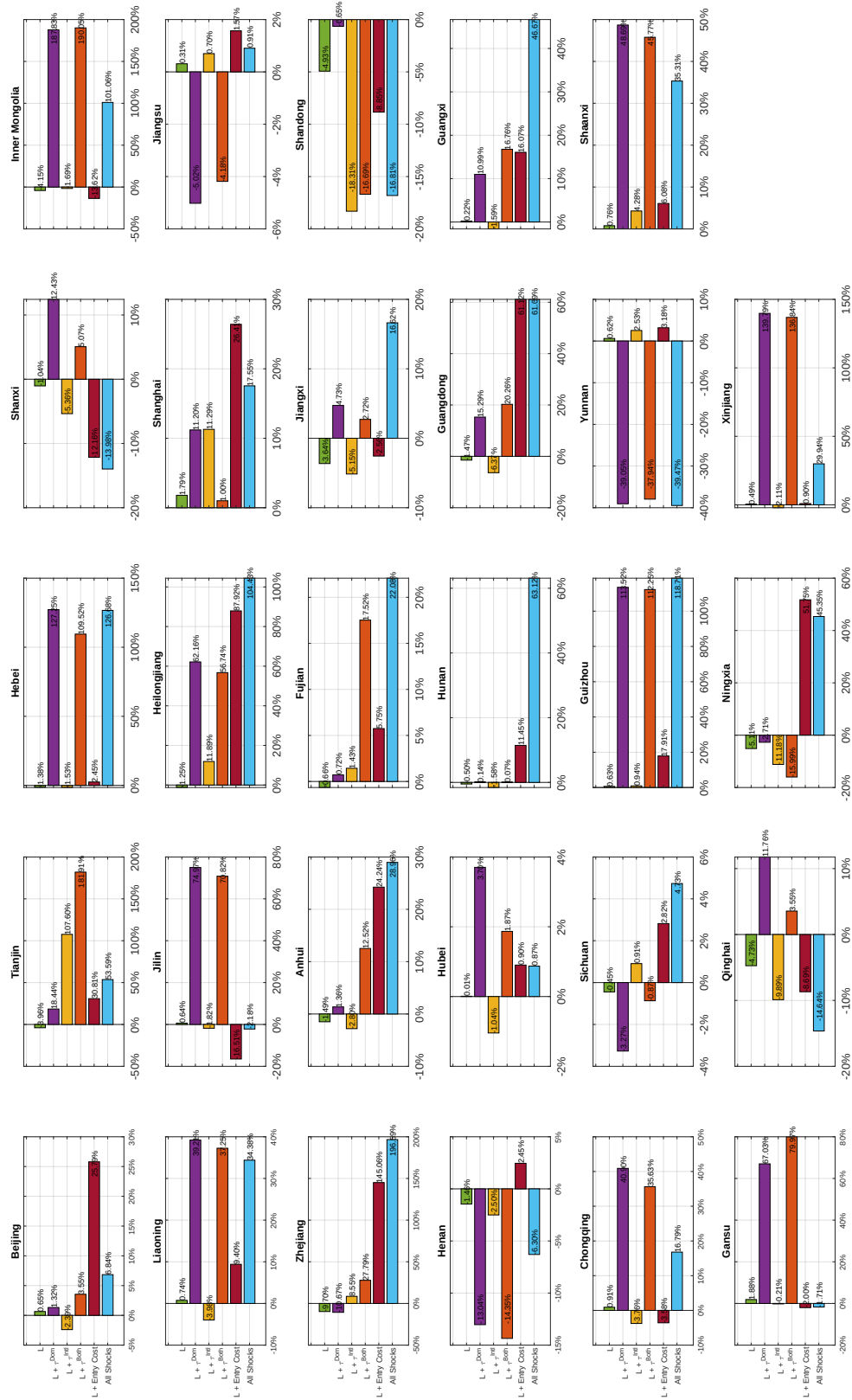


Figure A12: Change of Regional HHI